Linear mixed models
Part II

Marcello Gallucci
University of Milan-Bicocca

GLM

When the assumptions are NOT met because the data, and thus the errors, have more complex structures, we generalize the GLM to the Linear Mixed Model

Linear Mixed Model

GLM

Regression

T-test

ANOVA

ANCOVA

Moderation

Mediation

Path Analysis

LMM

Random coefficients models

Random intercept regression models

One-way ANOVA with random effects

One-way ANCOVA with random effects

Intercepts-and-slopes-as-outcomes models

Multi-level models

The mixed model

- We can now define a model with a regression for each cluster and the mean values of coefficients

Overall model

$$\hat{y}_{ij} = \bar{a} + a'_j + \bar{b} \cdot x_{ij} + b'_j \cdot x_{ij}$$

Random coefficients

Fixed coefficient

A GLM which contains both fixed and random effects is called a Linear Mixed Model

The mixed model

- In practice, mixed models allow to estimate the kind of effects we can estimate with the GLM, but they allow the effects to vary across clusters.
- Effects that vary across clusters are called **random effects**
- Effects that do not vary (the ones that are the same across clusters) are said to be **fixed effects**

Building a model

To build a model in a simple way, we need to answer very few questions:

- What is (are) the cluster variable(s)?
- What are the fixed effects?
- What are the random effects?

Repeated Measures Anova as a linear mixed model

A repeated measures design

- Consider now a classical repeated measures design (within-subjects) the levels of the WS IV (5 different conditions) are represented by different measures taken on the same person

Condition

		1	2	3	4	5
Participants	1	Y11	Y21	Y31	Y41	Y51
	2	Y12	Y22	Y32	Y42	Y52
	3	Y13	Y23	Y33	Y43	Y53
	….					
	N	Y1n	Y2n	Y3n	Y4n	Y5n

Standard file format

- As for many applications of the repeated-measure design, each level of the WS-factor is represented by a column in the file

One participant, one row

File Edit View Data Transform Analyze Direct Marketing Graphs Utilities Add-ons Window Help

1 : group

	group	err_t0	err_t1	err_t2	err_t3	err_t4	x
1	1	.14	.22	.439	.27	.01	-.04
2	1	.43	.52	.492	.48	.43	-.36
3	1	.61	.43	.446	.51	.57	-1.77
4	0	.29	.70	1.000	.89	.75	1.63
5	1	.16	.49	.500	.56	.29	-.32
6	0	.70	.36	.573	.57	.69	-1.16
7	0	.35	.51	.572	.46	.77	-.87
8	1	.45	.49	.545	.41	.43	-1.79
9	1	.05	.55	.333	.54	.53	1.01
10	1	.10	.35	.358	.57	.67	.58
11	0	.14	.45	.373	.25	.29	-.88
12	0	.04	.74	.541	.53	.35	-.27
13	1	.62	.73	.529	.31	.48	1.36
14	1	.15	.22	.101	.17	.17	-.32
15	0	.72	.55	.568	.53	.57	-.98

Long file format

- For the mixed model we need to tabulate the data as if they came from a between-subject design

One measure, one row

id	group	x	Condition	Y	va
1	1	-.037	1	.140	
1	1	-.037	2	.220	
1	1	-.037	3	.439	
1	1	-.037	4	.271	
1	1	-.037	5	.009	
2	1	-.360	1	.431	
2	1	-.360	2	.518	
2	1	-.360	3	.492	
2	1	-.360	4	.483	
2	1	-.360	5	.433	
3	1	-1.767	1	.612	
3	1	-1.767	2	.431	
3	1	-1.767	3	.446	
3	1	-1.767	4	.509	
3	1	-1.767	5	.573	
4	0	1.634	1	.291	
4	0	1.634	2	.702	

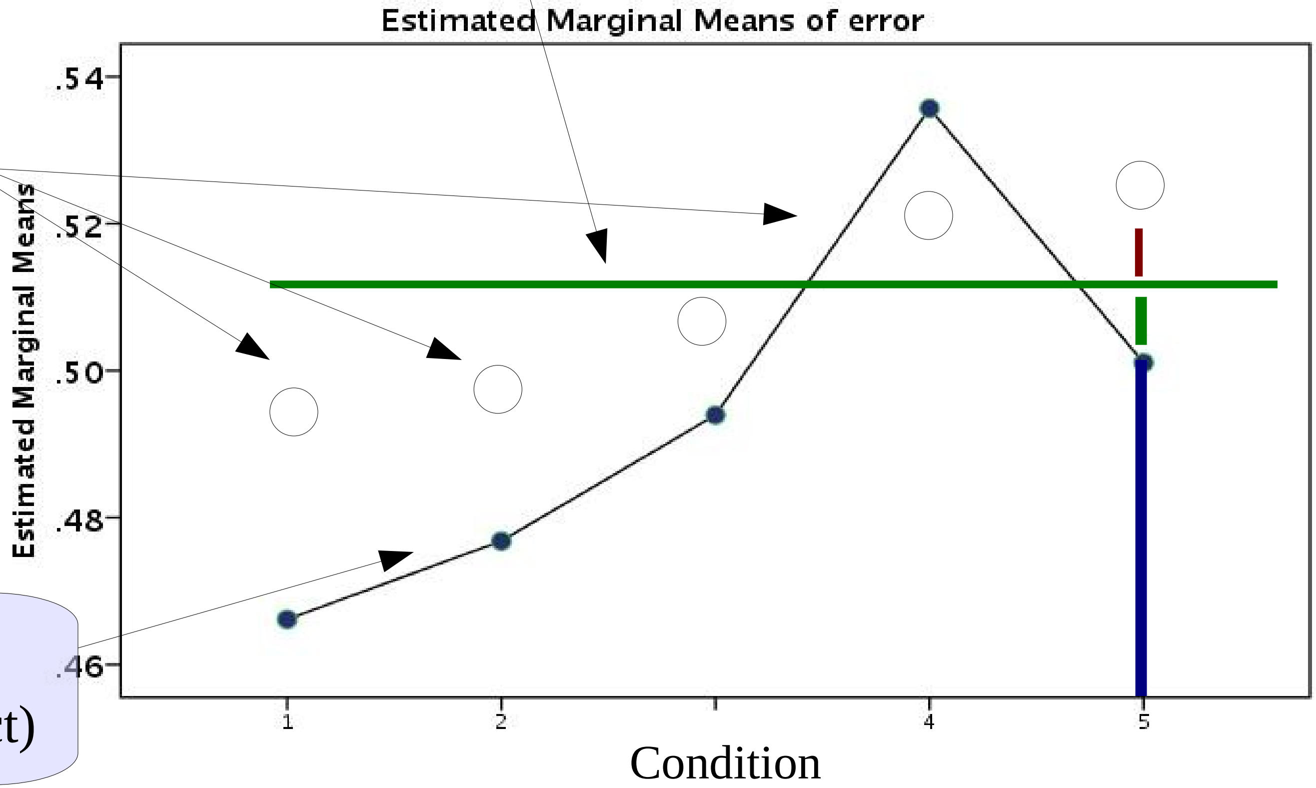
Participant scores

Plot for 1 participant

Participant average trait

Participant scores

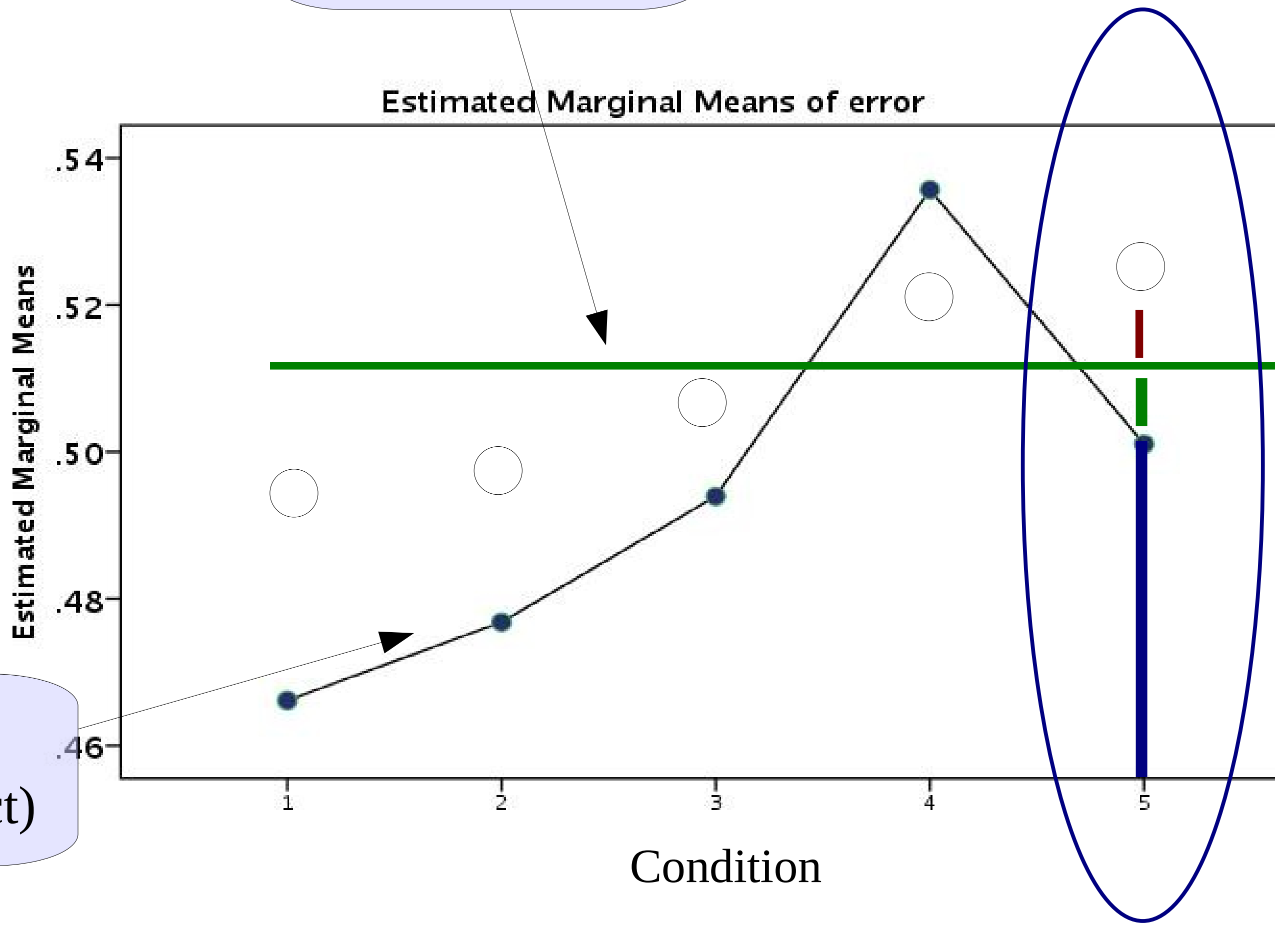
Averages of the sample (fixed effect)



Where does the score come from?

Plot for 1 participant

Participant average trait

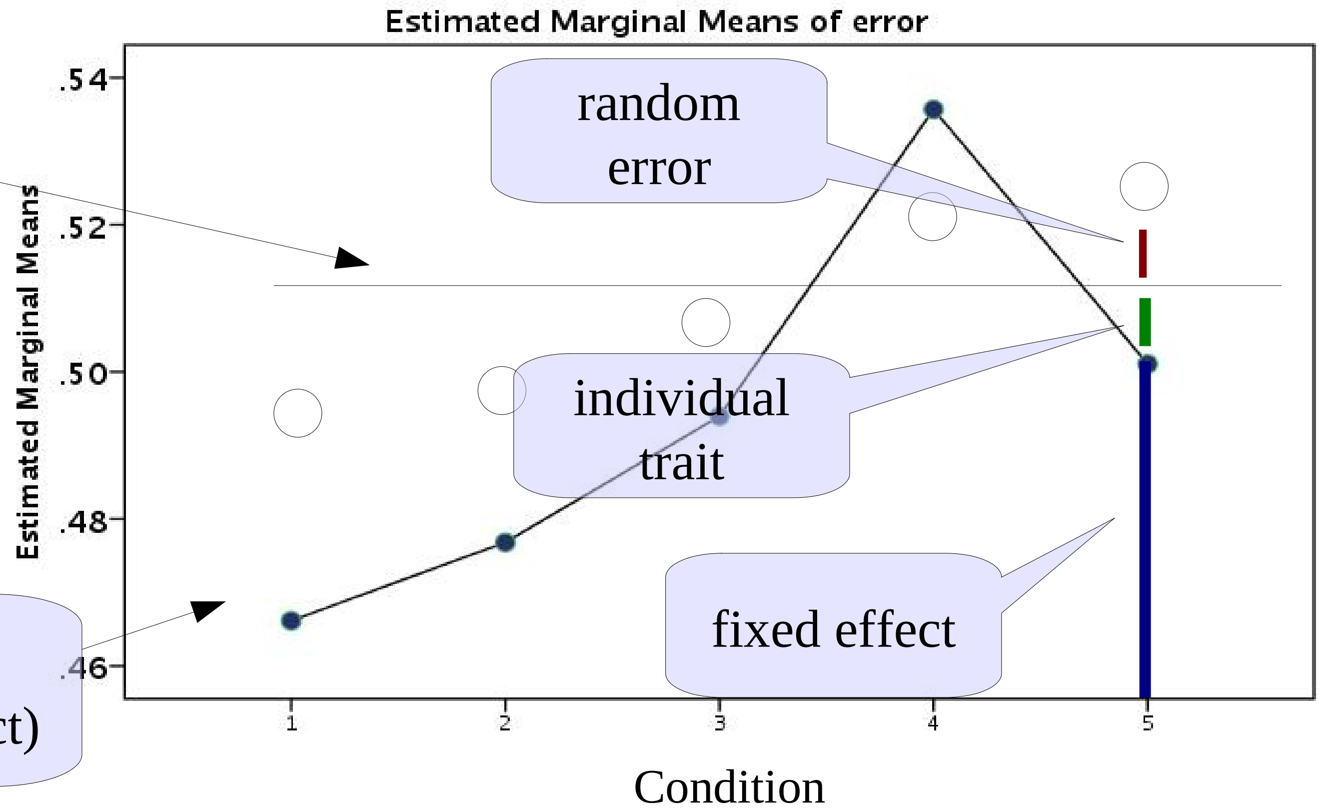


Averages of the sample (fixed effect)

Participant component

Plot for 1 participant

Participant individual trait



Averages of the sample (fixed effect)

Solution

Thus, we should consider an extra residual term which represents participants individual characteristic. This term is the same within each participant

one participant
one trait

$$Y_{11} = a + b_1 \cdot T_1 + u_1 + e_{11}$$
$$Y_{21} = a + b_2 \cdot T_2 + u_1 + e_{21}$$
$$Y_{31} = a + b_3 \cdot T_3 + u_1 + e_{31}$$

Each score,
one residual

Effects of
condition

$$Y_{1j} = a + b_1 \cdot T_1 + u_j + e_{1j}$$
$$Y_{2j} = a + b_2 \cdot T_2 + u_j + e_{2j}$$
$$Y_{3j} = a + b_3 \cdot T_3 + u_j + e_{3j}$$

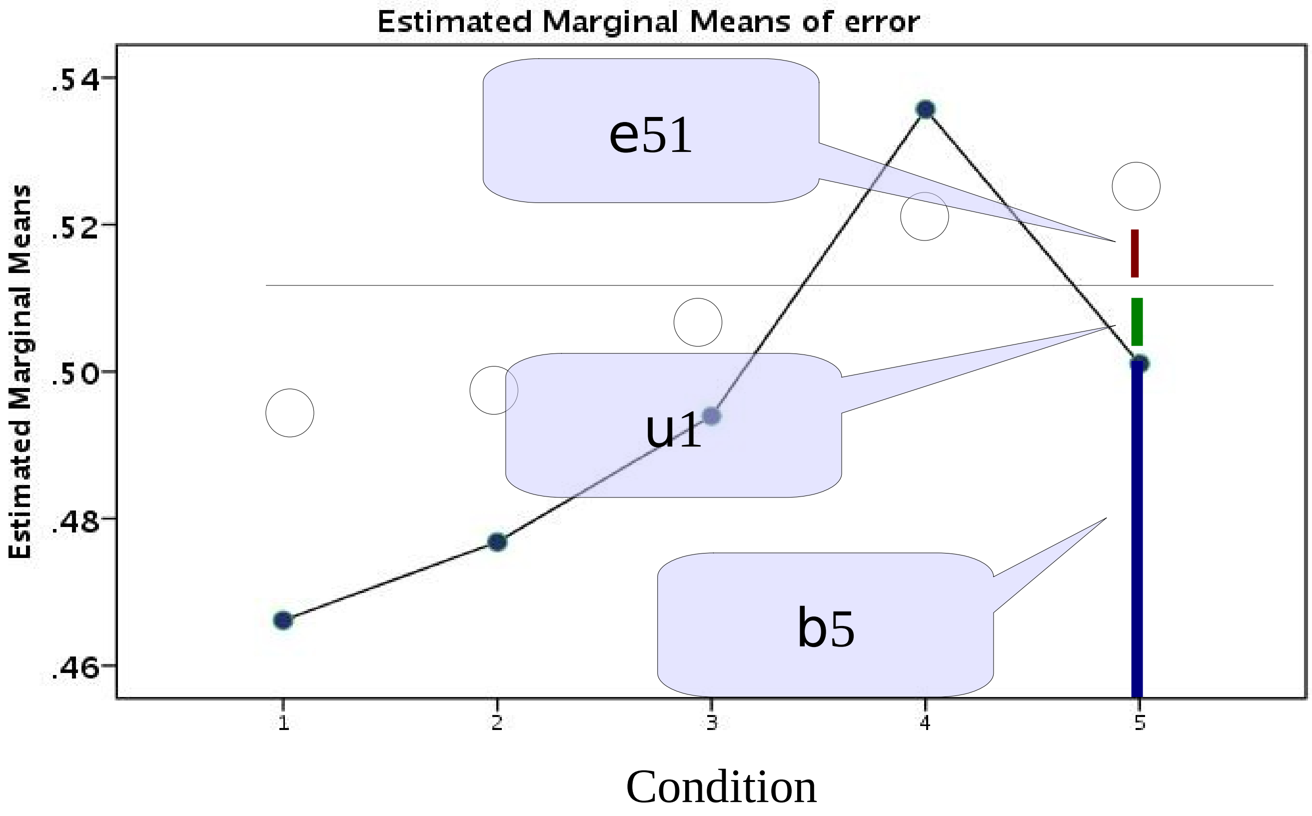
Each score,
one error

One participant
one trait

We assume the 5 conditions are dummy coded

Participant component

$$Y_{51} = a + b \cdot T_5 + u_1 + e_{51}$$



Building the model

We translate this in the standard mixed model

$$Y_{ij} = a + b' \cdot T_i + u_j + e_{ij}$$

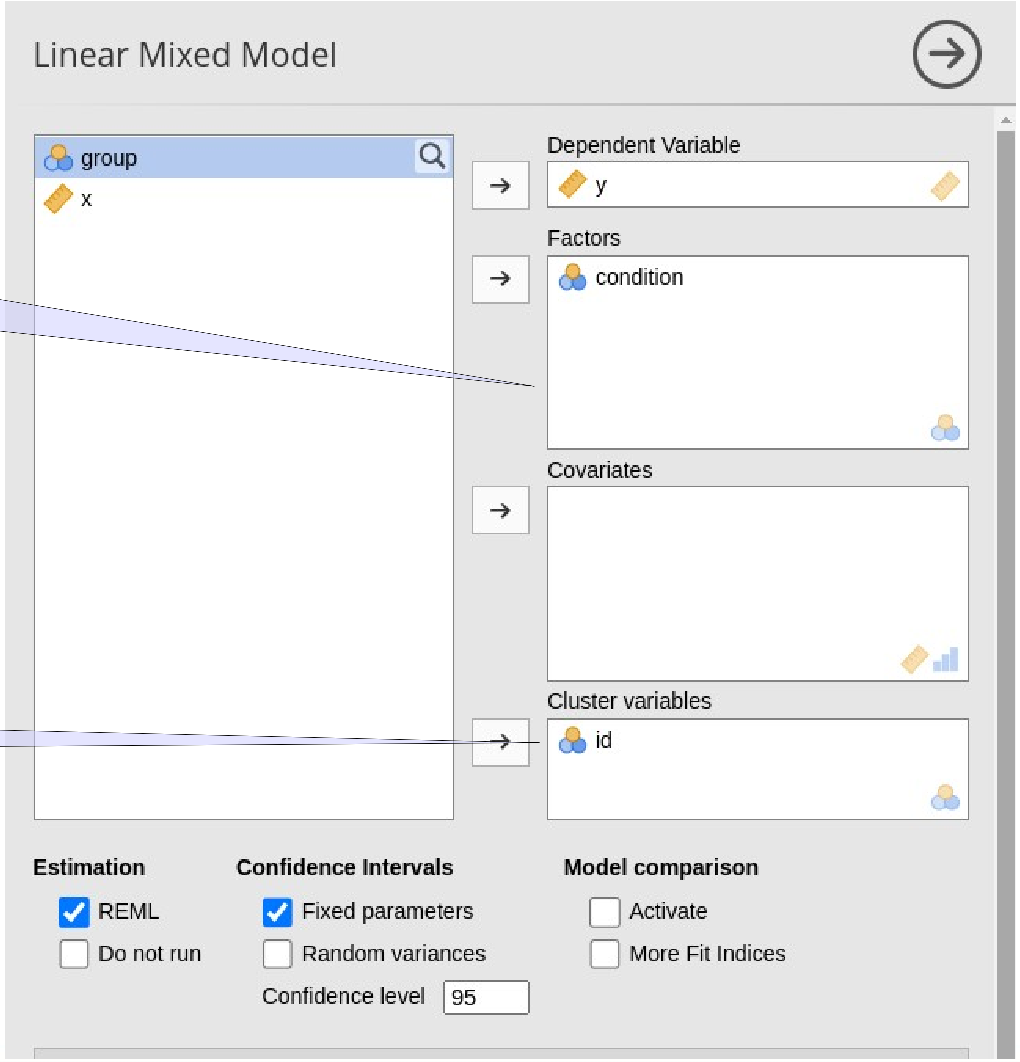
$$y_{ij} = \bar{a} + a_j + \bar{b} \cdot x_{ij} + e_{ij}$$

- Fixed effects? Intercept and condition effect
- Random effects? Intercepts
- Clusters? participants

GAMLj: General mixed models

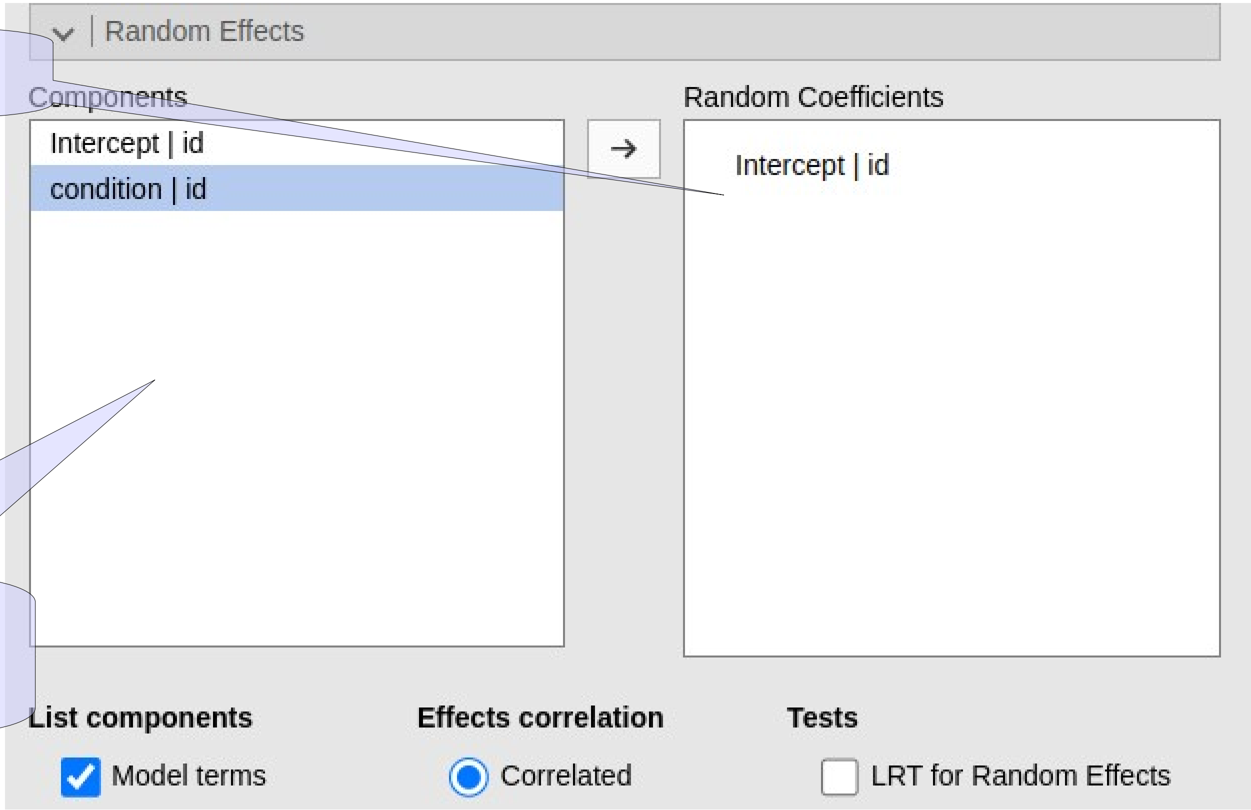
Categorical independent variable

Clustering variable(s)



GAMLj: random coefficients

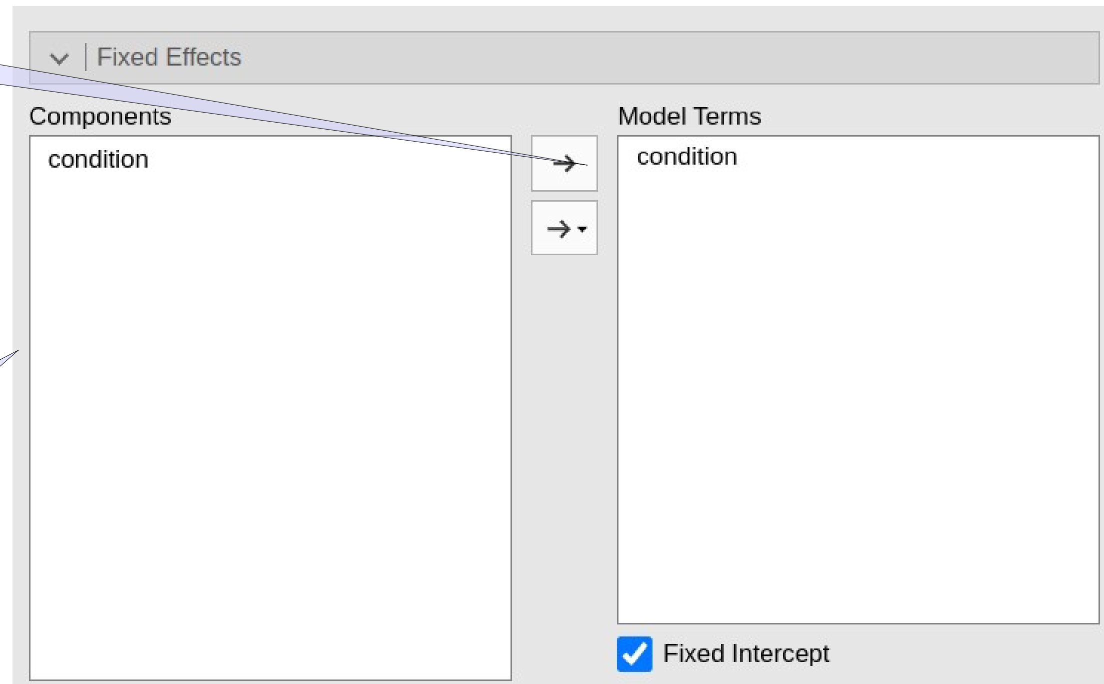
Random intercepts



All possible random coefficients

GAMLj: fixed coefficients

Fixed effect



All possible fixed coefficients

GAMLj: Results: model

R-squared

Model Results

Model Fit

Type	R^2	df	LRT X^2	p
Conditional	0.217	5	78.931	< .001
Marginal	0.015	4	18.773	< .001

[4]

R-squared Conditional: How much variance can the fixed and random effects together explain of the overall variance

R-squared Marginal: How much variance can the fixed effects alone explain of the overall variance

GAMLj: Results: random

Variance of intercepts

Random Components

Groups	Name	Variance	SD	ICC
id	(Intercept)	0.00780	0.0883	0.205
Residual		0.03020	0.1738	

Note. Number of Obs: 1000 , Number of groups: id 200

As long as the variance is non-zero, we are fine

GAMLj: Results: fixed

F-tests

Fixed Effects Omnibus Tests

	F	df	df (res)	p
condition	4.72	4	796	< .001

Coefficients

Parameter Estimates (Fixed coefficients)

				95% Confidence Intervals				
Names	Effect	Estimate	SE	Lower	Upper	df	t	p
(Intercept)	(Intercept)	0.4947	0.00832	0.47841	0.5111	199	59.462	< .001
condition1	2 - 1	0.0107	0.01738	−0.02344	0.0448	796	0.613	0.540
condition2	3 - 1	0.0278	0.01738	−0.00632	0.0619	796	1.598	0.110
condition3	4 - 1	0.0695	0.01738	0.03541	0.1036	796	4.000	< .001
condition4	5 - 1	0.0349	0.01738	8.03e−4	0.0690	796	2.009	0.045

Contrasts used to cast the categorical IV

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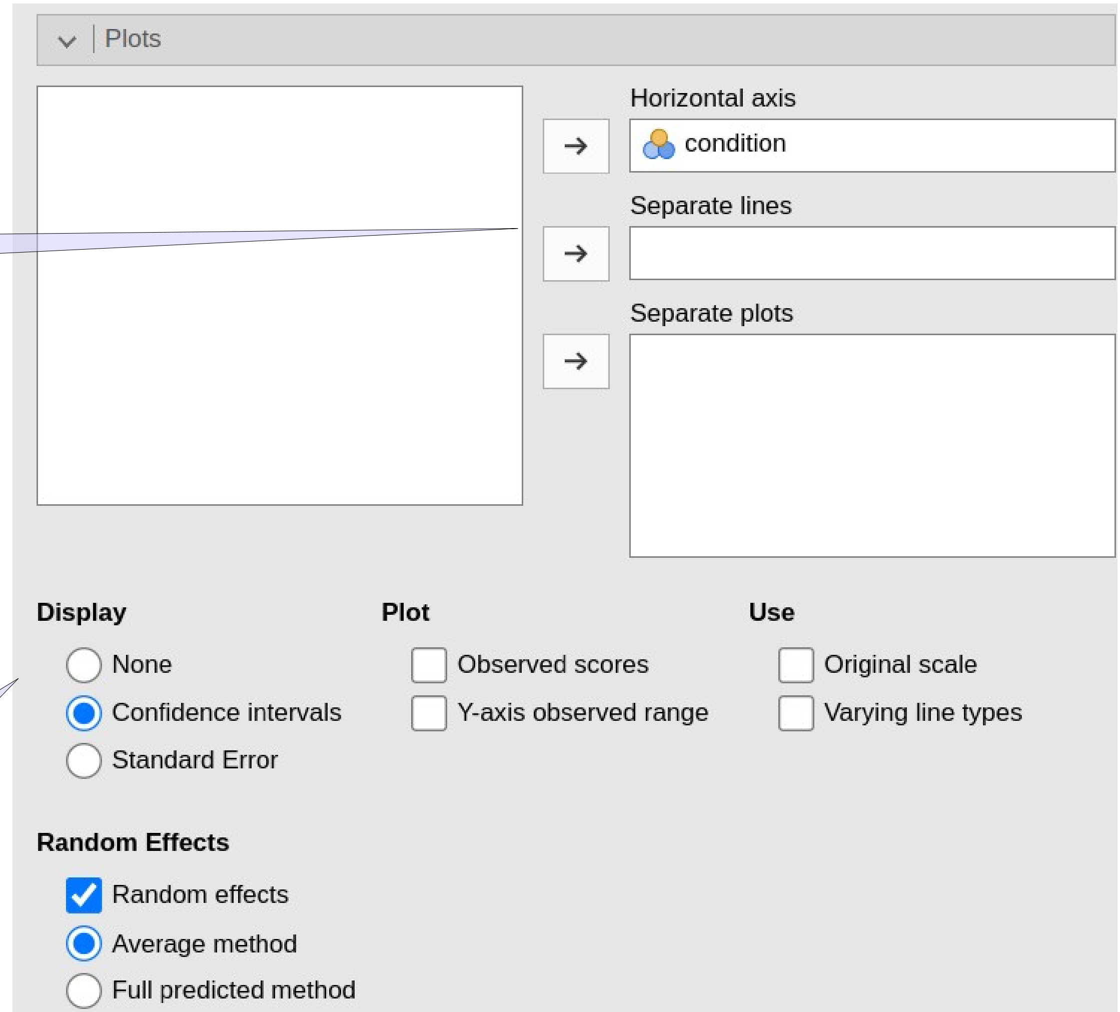
GAMLj: plot

Fixed effect to plot

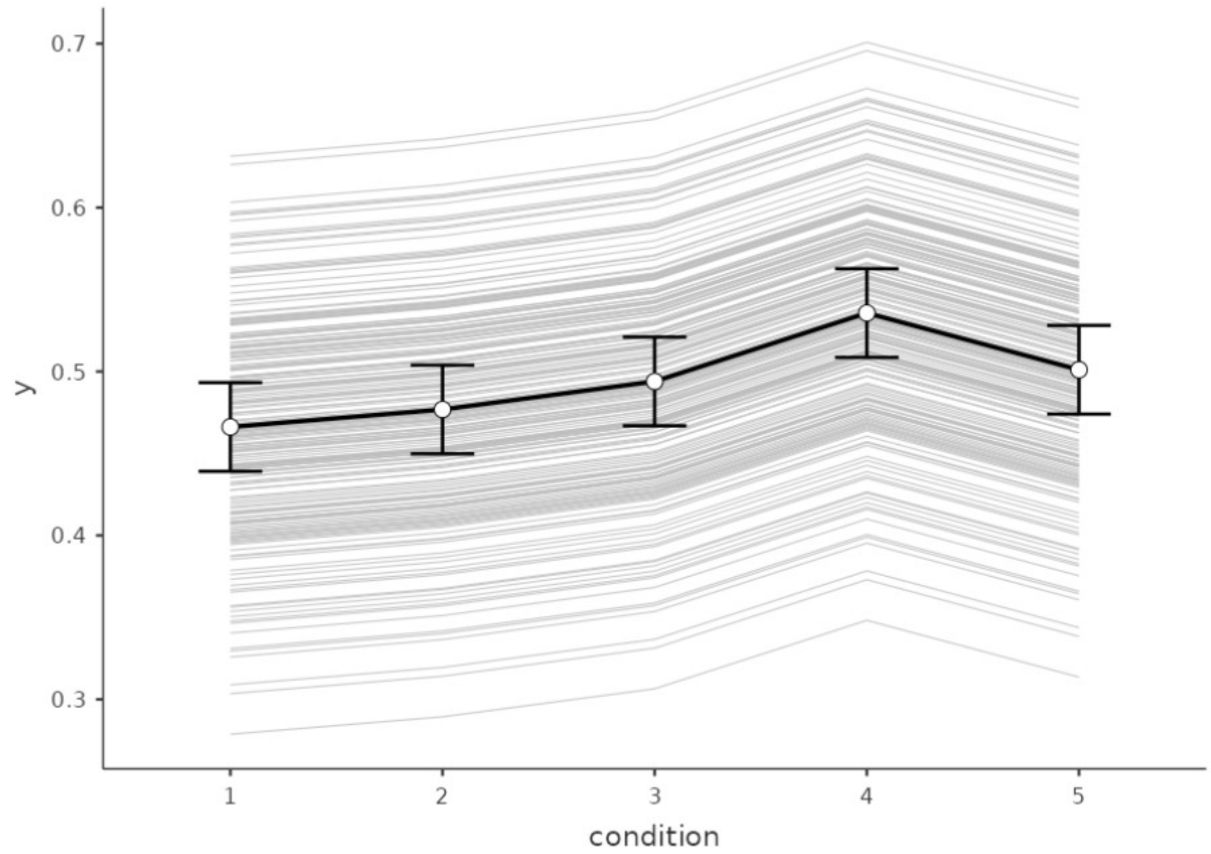


Options

GAMLj: plot

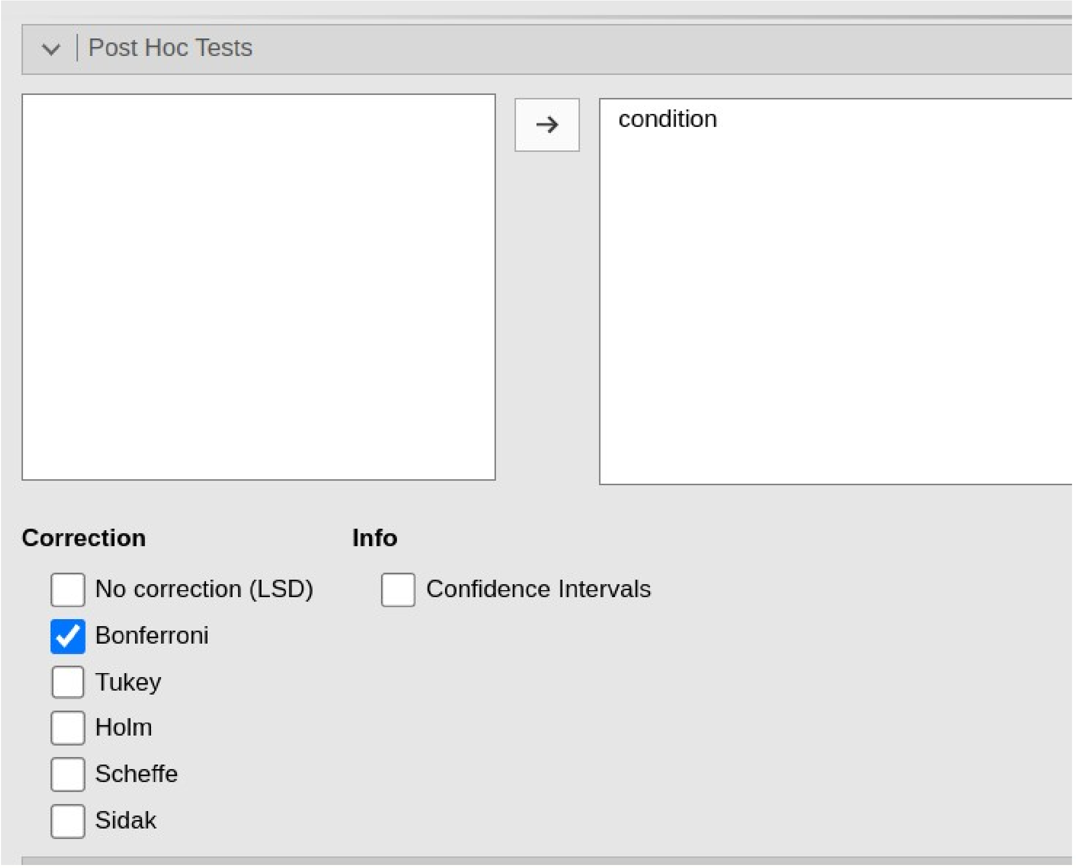
Results Plots

condition



GAMLj: post-hoc

- As in GLM (Anova), sometimes we want to compares conditions using post-hoc tests. GAMLj allows for Bonferroni and Tukey (more liberal) p-value adjustement



GAMLj: post-hoc

- The interpretation follows as for any standard ANOVA

Post Hoc Tests

Post Hoc Comparisons - trial

Comparison								
trial		trial	Difference	SE	t	df	$p_{bonferroni}$	p_{holm}
1	-	2	−0.01066	0.0174	−0.613	796	1.000	1.000
	-	3	−0.02778	0.0174	−1.598	796	1.000	0.552
	-	4	−0.06951	0.0174	−4.000	796	< .001	< .001
	-	5	−0.03491	0.0174	−2.009	796	0.449	0.314
2	-	3	−0.01712	0.0174	−0.985	796	1.000	0.975
	-	4	−0.05885	0.0174	−3.386	796	0.007	0.007
	-	5	−0.02425	0.0174	−1.395	796	1.000	0.653
3	-	4	−0.04173	0.0174	−2.401	796	0.166	0.133
	-	5	−0.00713	0.0174	−0.410	796	1.000	1.000
4	-	5	0.03460	0.0174	1.991	796	0.468	0.314

Between and Repeated Measures Anova

linear mixed model

Standard design

- There are two groups - a Control group and a Treatment group, measured at 4 times. These times are labeled as 1 (pretest), 2 (one month posttest), 3 (3 months follow-up), and 4 (6 months follow-up).

- The dependent variable is a depression score (e.g. Beck Depression Inventory) and the treatment is drug versus no drug. If the drug worked about as well for all subjects the slopes would be comparable and negative across time. For the control group we would expect some subjects to get better on their own and some to stay depressed, which would lead to differences in slope for that group (*)

*) https://www.uvm.edu/~dhowell/StatPages/More_Stuff/Mixed-Models-Repeated/Mixed-Models-for-Repeated-Measures1.html

Standard design

- There are two groups - a Control group and a Treatment group, measured at 4 times. These times are labeled as 1 (pretest), 2 (one month posttest), 3 (3 months follow-up), and 4 (6 months follow-up).

96 observations
24 subjects

Contingency Tables

Contingency Tables

	group		
time	1	2	Total
0	12	12	24
1	12	12	24
3	12	12	24
6	12	12	24
Total	48	48	96

*) https://www.uvm.edu/~dhowell/StatPages/More_Stuff/Mixed-Models-Repeated/Mixed-Models-for-Repeated-Measures1.html

Standard design: data

- Data are in the long format

One subject 4 rows

	subj	time	group	dv
1	1	0	1	296
2	1	1	1	175
3	1	3	1	187
4	1	6	1	192
5	2	0	1	376
6	2	1	1	329
7	2	3	1	236
8	2	6	1	76
9	3	0	1	309
10	3	1	1	238
11	3	3	1	150
12	3	6	1	123
13	4	0	1	222
14	4	1	1	60
15	4	3	1	82
16	4	6	1	85
17	5	0	1	150
18	5	1	1	271
19	5	3	1	250

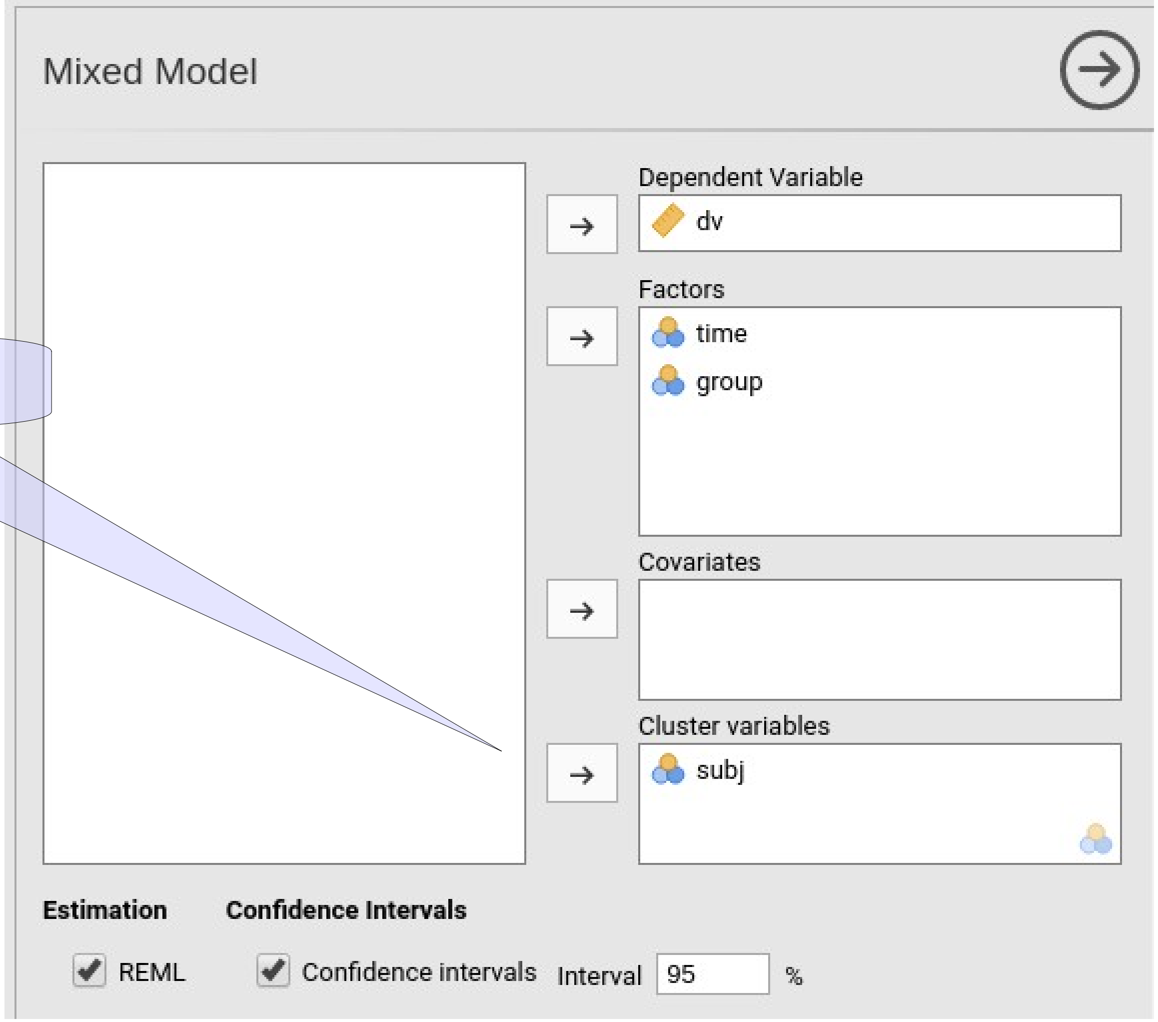
Mixed model

We can translate this in a standard mixed model

- Fixed effects? Intercept and group,time, and interaction effect
- Random effects? Intercepts
- Clusters? subjects

Variables

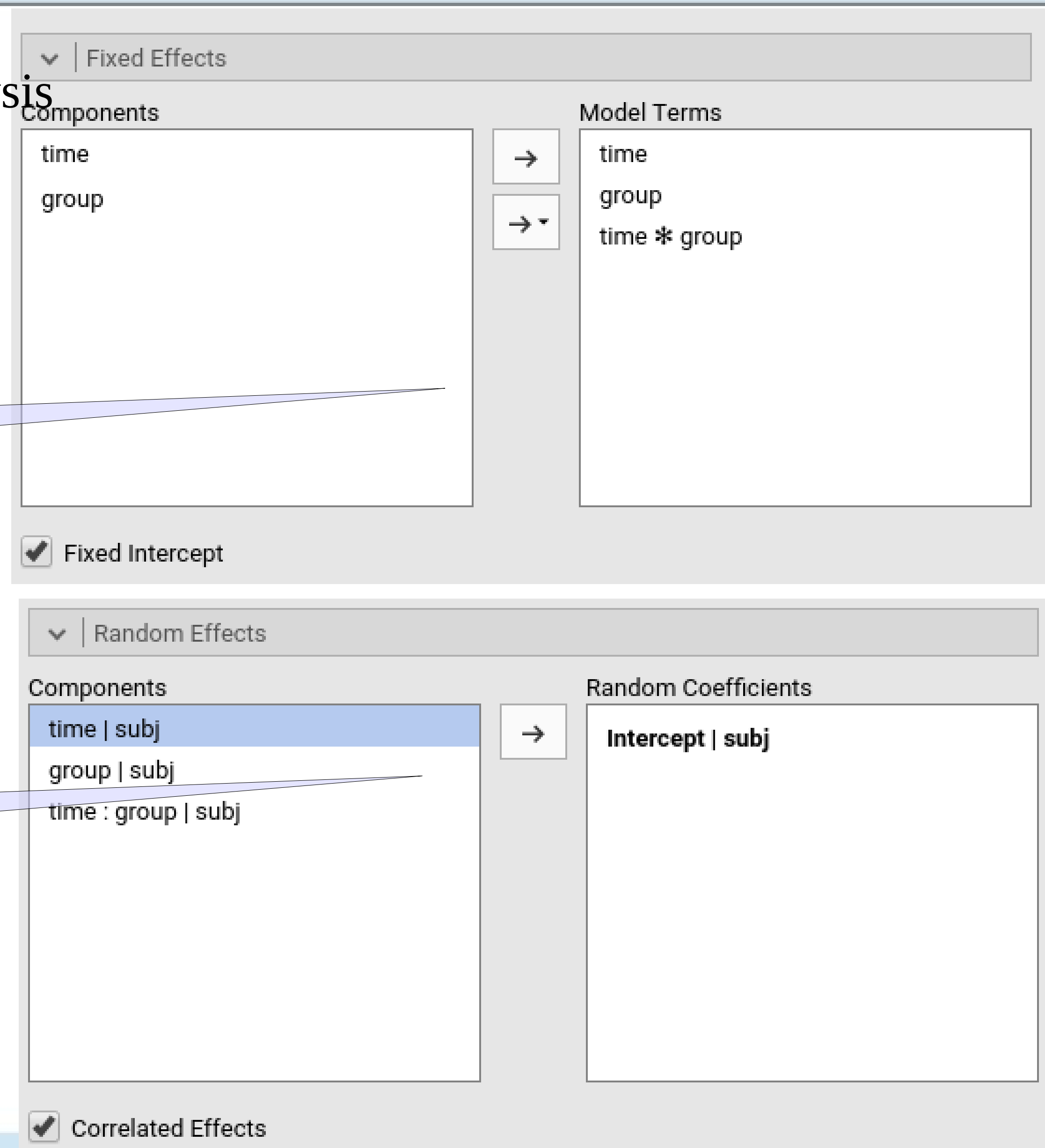
- Definition of the analysis



Clustering variable

Model

- Definition of the analysis



Fixed effects

Random effects

Results

- Interpretation of results

Model

Model Results

Model Fit

Type	R^2	df	LRT X^2	p
Conditional	0.768	8	107.251	< .001
Marginal	0.554	7	101.043	< .001

[4]

Random effects

Random Components

Groups	Name	Variance	SD	ICC
subj	(Intercept)	2539	50.4	0.479
Residual		2761	52.5	

Note. Number of Obs: 96 , Number of groups: subj 24

Results

- Interpretation of results

Fixed F-tests

Fixed Effect ANOVA

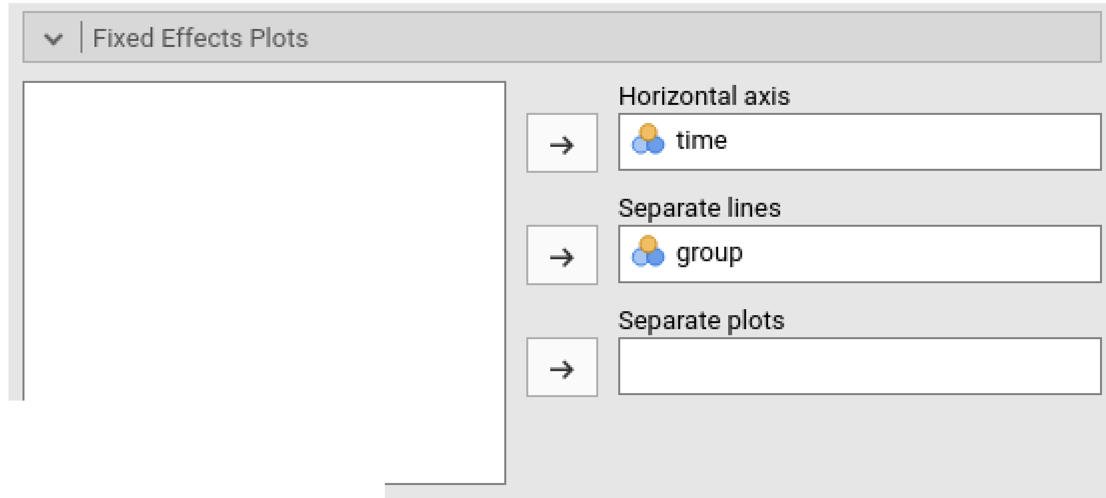
	F	Num df	Den df	p
time	45.14	3	66.0	< .001
group	13.71	1	22.0	0.001
time:group	9.01	3	66.0	< .001

Note. Satterthwaite method for degrees of freedom

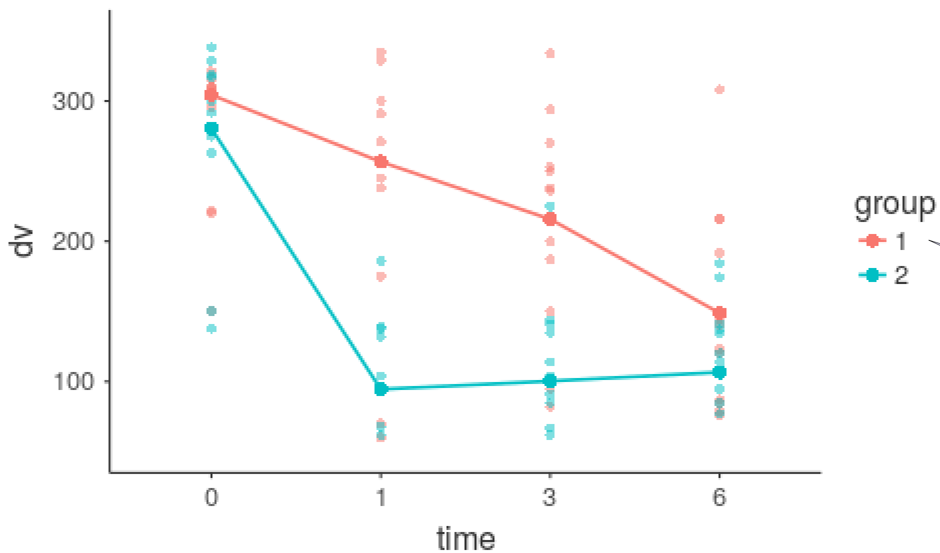
- For the moment we ignore the coefficients of the parameter estimates

Results: plot

- Interpretation of results



Fixed Effects Plots



Red is the control group

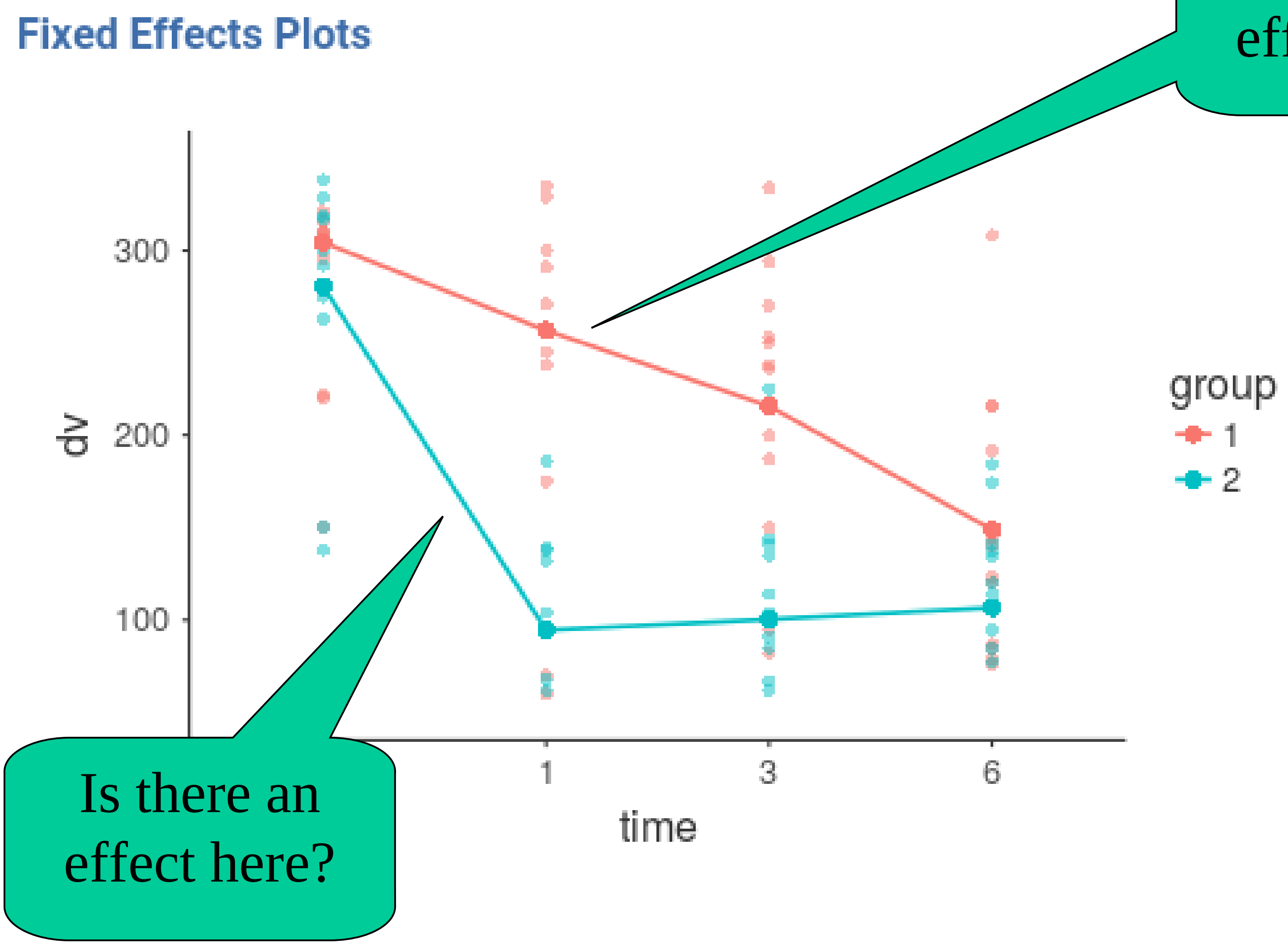
Probing the results

- We can probe the interaction (and the pattern of means) in different ways (all available in GAMLj):
- Simple effects: Test if the effects of time is there (and how strong it is) for different groups
- Trend analysis: Checking the polynomial trend for time in general and for different groups
- Post-hoc test

Simple effect analysis

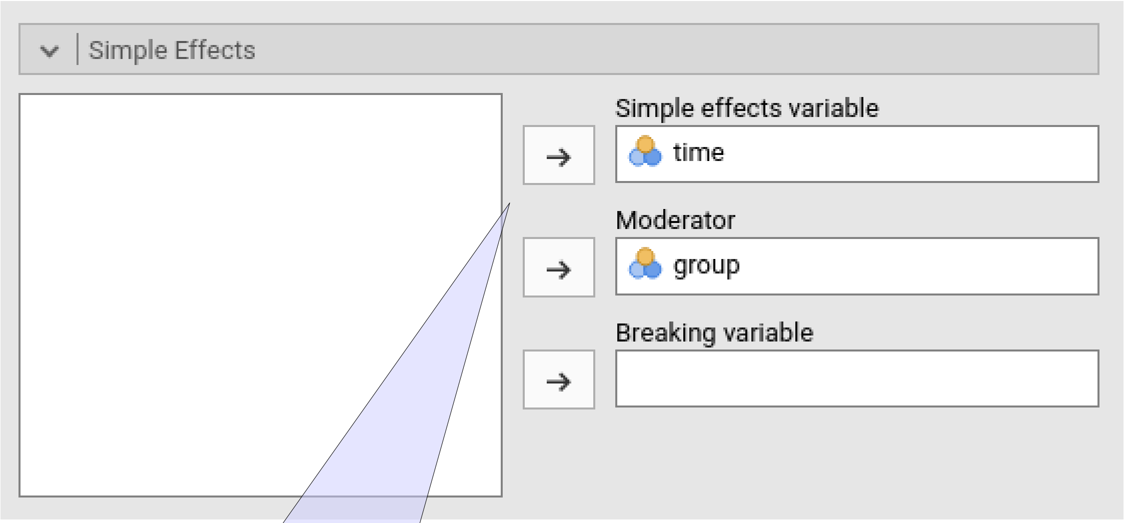
Simple effects

- Interpretation of results



Simple effects

- We should declare which is the variable we want the effect for and which is the moderator



Effects of time for different groups

Simple effects

- We can say that the treatment works for both groups, although in a different way (recall the interaction)

Simple Effects ANOVA

Simple effects of time

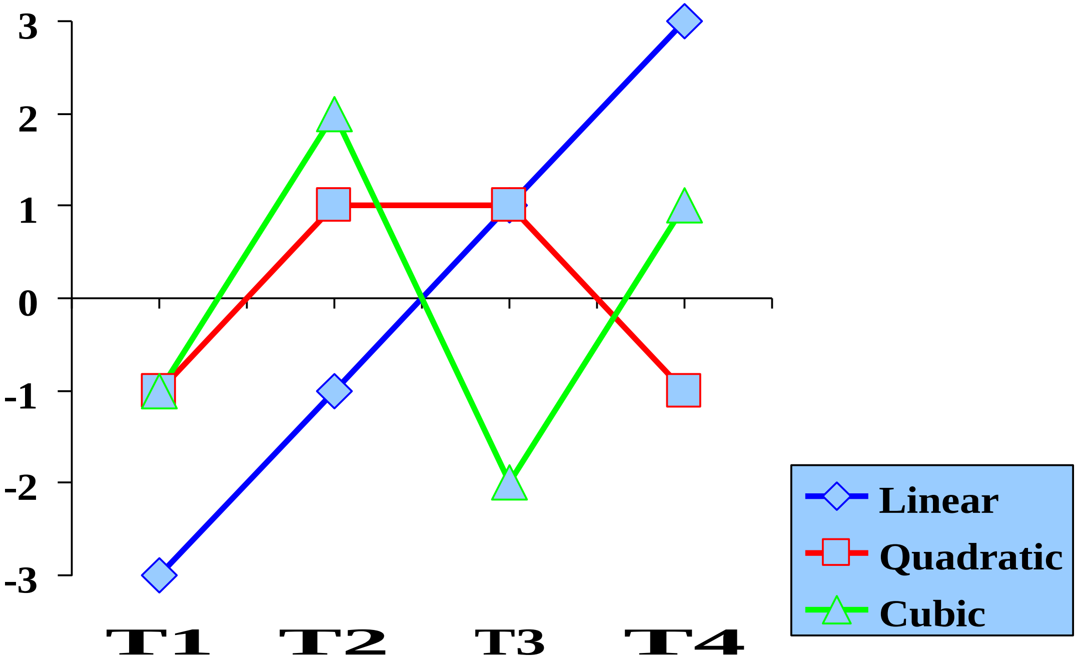
Effect	Moderator Levels	df Num	df Den	F	p
time	group at 1	3.00	66.0	18.9	< .001
time	group at 2	3.00	66.0	35.3	< .001

In both groups there is an affect of time

Trend analysis

Polynomial Contrasts

Trend analysis is based on Polynomial contrasts: each contrast features weights which follow well-known shapes (polynomial functions)



$$L = \begin{pmatrix} -3 & -1 & 1 & 3 \end{pmatrix}$$

$$Q = \begin{pmatrix} -1 & 1 & 1 & -1 \end{pmatrix}$$

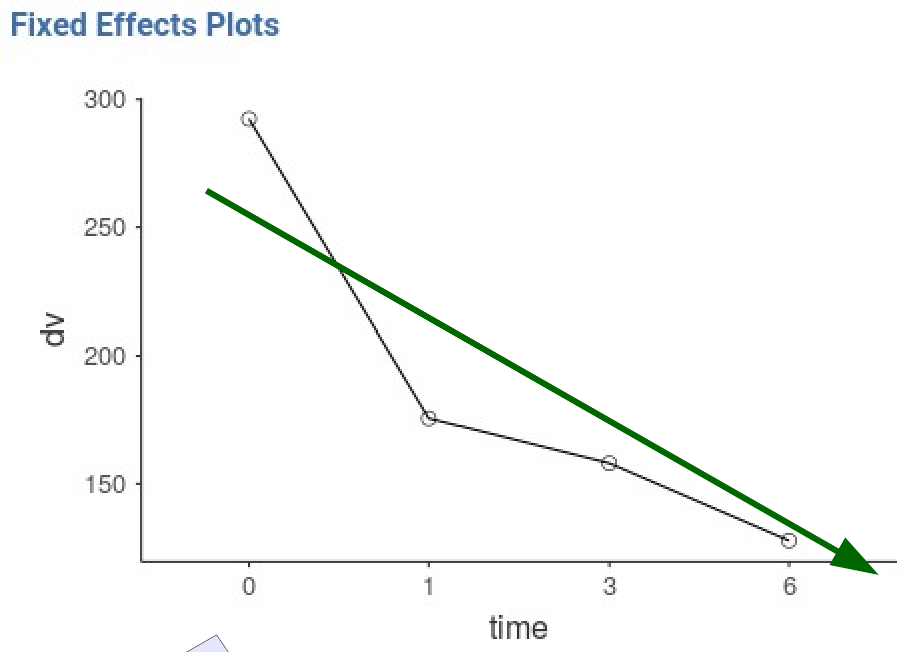
$$C = \begin{pmatrix} -1 & 2 & -2 & 1 \end{pmatrix}$$

Trend analysis

- It is useful to test what kind of trend is present in the pattern of means
- It can be applied to any ordered categorical variables
- It is often used (and SPSS gives it by default) in repeated measures analysis
- One can estimate K-1 trends (linear, quadratic, cubic etc), where K is the number of means (conditions)

Trend analysis

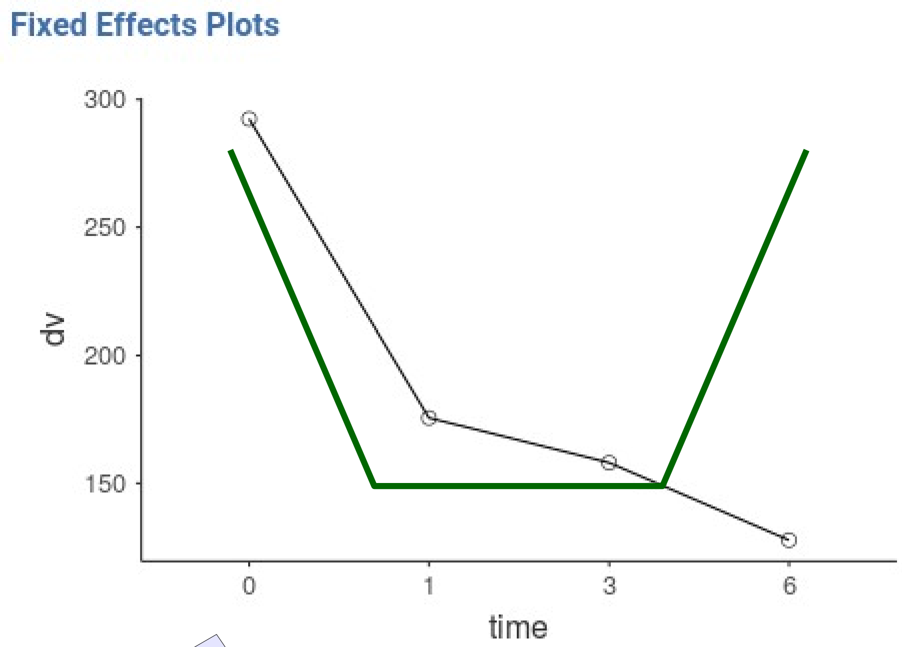
- Each trend (linear, quadratic, etc) tests a particular shape of the mean pattern



Linear: on average means go down (or up, not flat)

Trend analysis

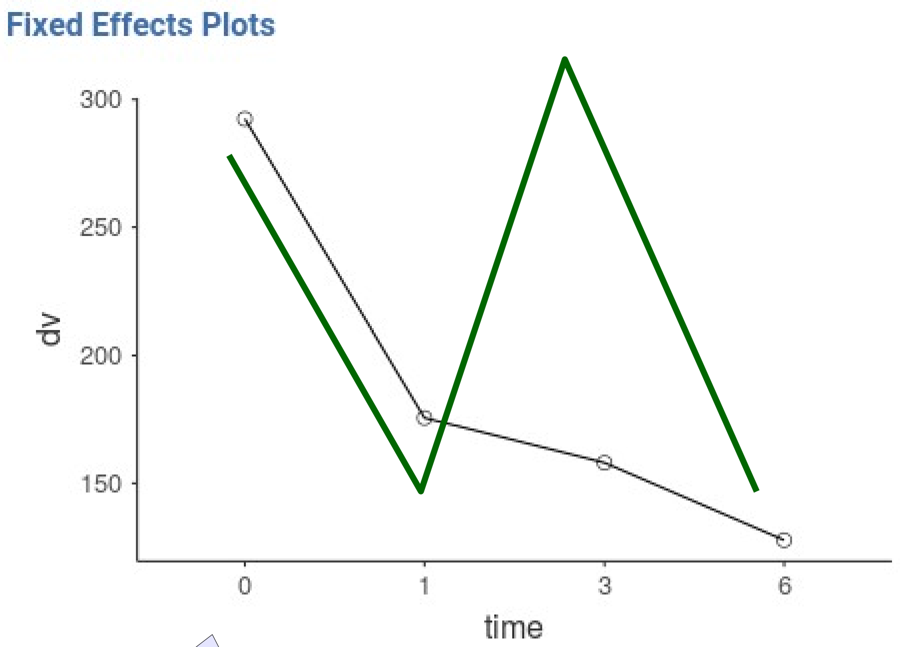
- Each trend (linear, quadratic, etc) tests a particular shape of the mean pattern



Quadratic: on average means go down and then up

Trend analysis

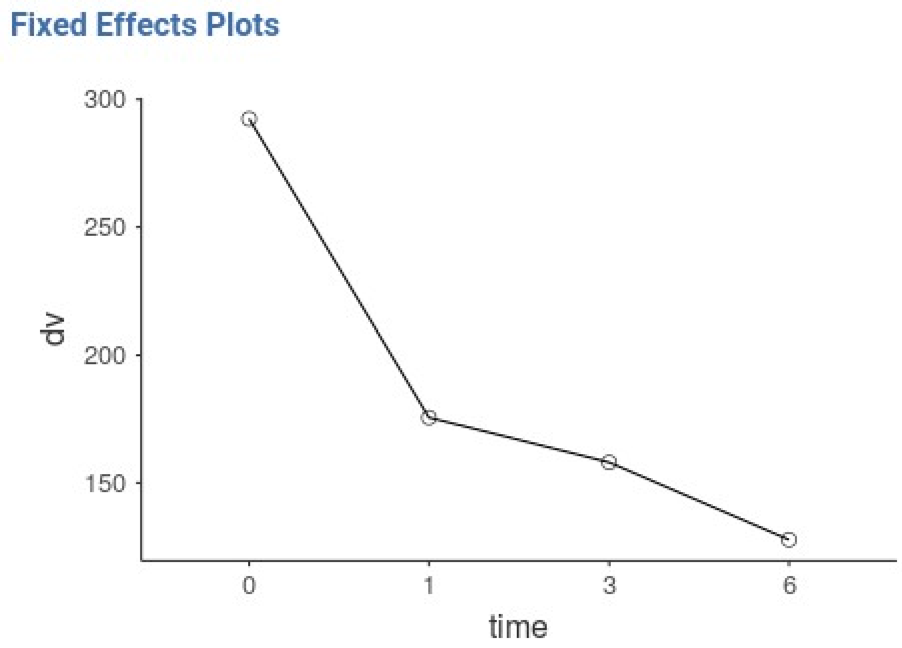
- Each trend (linear, quadratic, etc) tests a particular shape of the mean pattern



Cubic: on average means fluctuate

Trend analysis

- Each significant trend justifying interpreting a particular characteristic of the mean pattern



GAMLj:Trend analysis

- First, we should code the categorical variable “time” as a polynomial contrast

- We can leave “group” as simple (default) which means “centered dummy contrasts”

GAMLj:Trend analysis

- Second, look at the parameter estimates

Contrast labels

Contrast average effects

Parameter Estimates (Fixed coefficients)

Names	Effect	Estimate	SE	95% Confidence Intervals Lower	Upper	df	t	p
(Intercept)	(Intercept)	188.44	11.6	165.4	211.50	22.0	16.2444	< .001
time1	linear	−114.36	10.7	−135.7	−93.04	66.0	−10.6626	< .001
time2	quadratic	43.25	10.7	21.9	64.57	66.0	4.0326	< .001
time3	cubic	−25.04	10.7	−46.4	−3.72	66.0	−2.3351	0.023
group1	2 - 1	−85.92	23.2	−132.0	−39.80	22.0	−3.7033	0.001
time1 ✻ group1	(linear) ✻ (2 - 1)	−1.79	21.5	−44.4	40.85	66.0	−0.0834	0.934
time2 ✻ group1	(quadratic) ✻ (2 - 1)	105.75	21.5	63.1	148.39	66.0	4.9301	< .001
time3 ✻ group1	(cubic) ✻ (2 - 1)	−35.44	21.5	−78.1	7.20	66.0	−1.6523	0.103

[5]

Contrast interaction with group

GAMLj:Trend analysis

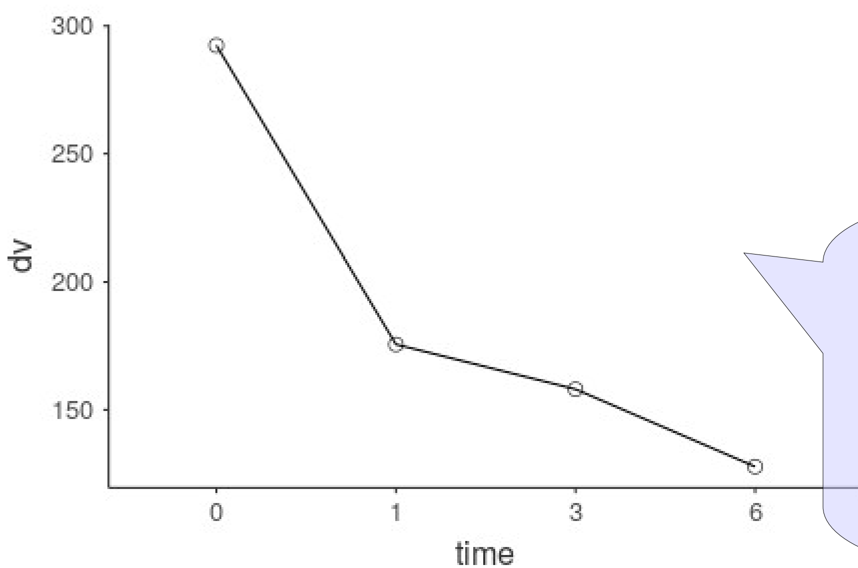
- Average effects of the contrasts

Parameter Estimates (Fixed coefficients)

Names	Effect	Estimate	SE	95% Confidence Intervals Lower	Upper	df	t	p
(Intercept)	(Intercept)	188.44	11.6	165.4	211.50	22.0	16.2444	< .001
time1	linear	−114.36	10.7	−135.7	−93.04	66.0	−10.6626	< .001
time2	quadratic	43.25	10.7	21.9	64.57	66.0	4.0326	< .001
time3	cubic	−25.04	10.7	−46.4	−3.72	66.0	−2.3351	0.023
				[illegible]	−39.80	22.0	−3.7033	0.001
				[illegible]	40.85	66.0	−0.0834	0.934
				[illegible]	148.39	66.0	4.9301	< .001
				[illegible]	7.20	66.0	−1.6523	0.103

[5]

Fixed Effects Plots



The pattern (on average) shows all three trends:
1. it goes down (linear)
2. it tend to go down and then up
3. if fluctuates a bit

GAMLj:Trend analysis

- Trend analysis by group

Parameter Estimates (Fixed coefficients)

				95% Confidence Intervals				
Names	Effect	Estimate	SE	Lower	Upper	df	t	p
(Intercept)	(Intercept)	188.44	11.6	165.4	211.50	22.0	16.2444	< .001
time1	linear	−114.36	10.7	−135.7	−93.04	66.0	−10.6626	< .001
time2	quadratic	43.25	10.7	21.9	64.57	66.0	4.0326	< .001
time3	cubic	−25.04	10.7	−46.4	−3.72	66.0	−2.3351	0.023
group1	2 - 1	−85.92	23.2	−132.0	−39.80	22.0	−3.7033	0.001
time1 ✻ group1	(linear) ✻ (2 - 1)	−1.79	21.5	−44.4	40.85	66.0	−0.0834	0.934
time2 ✻ group1	(quadratic) ✻ (2 - 1)	105.75	21.5	63.1	148.39	66.0	4.9301	< .001
time3 ✻ group1	(cubic) ✻ (2 - 1)	−35.44	21.5	−78.1	7.20	66.0	−1.6523	0.103

[5]

Those tell us if the trend is
different between the two groups:
Linear: no
Quadratic: yes
Cubib: no

Both groups decreases
Group 2 curve is stronger
They both fluctuates a bit

GAMLj:Trend analysis

- Trend analysis by group

Parameter Estimates (Fixed coefficients)

Names	Effect	Estimate	SE	95% Confidence Intervals Lower	Upper	df	t	p
(Intercept)	(Intercept)	188.44	11.6	165.4	211.50	22.0	16.2444	< .001
time1	linear	−114.36	10.7	−135.7	−93.04	66.0	−10.6626	< .001
time2	quadratic	43.25	10.7	21.9	64.57	66.0	4.0326	< .001
time3	cubic	−25.04	10.7	−46.4	−3.72	66.0	−2.3351	0.023
group1	2 - 1	−85.92	23.2	−132.0	−39.80	22.0	−3.7033	0.001
time1 ✻ group1	(linear) ✻ (2 - 1)	−1.79	21.5	−44.4	40.85	66.0	−0.0834	0.934
time2 ✻ group1	(quadratic) ✻ (2 - 1)	105.75	21.5	63.1	148.39	66.0	4.9301	< .001
time3 ✻ group1	(cubic) ✻ (2 - 1)	−35.44	21.5	−78.1	7.20	66.0	−1.6523	0.103

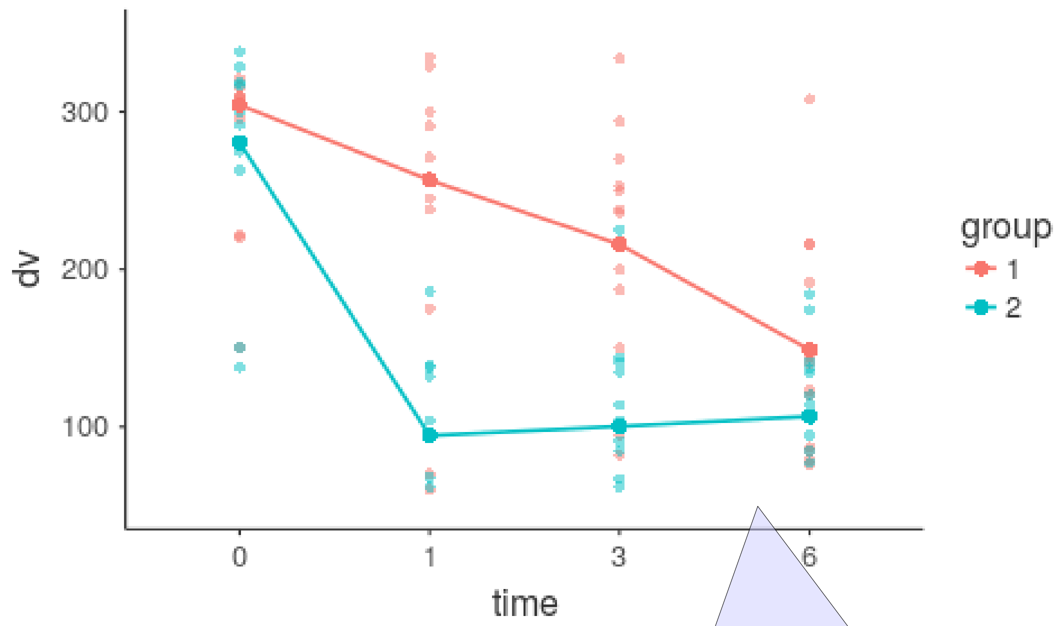
[5]

Those tell us if the trend is
different between the two groups:
Linear: no
Quadratic: yes
Cubic: mild

Both groups decreases
One group has a stronger curve
They both fluctuates a bit

GAMLj:Trend analysis

Fixed Effects Plots



Those tell us if the trend is different between the two groups:
Linear: no
Quadratic: yes
Cubic: mild

Both groups decreases
Group 2 curve is stronger
The fluctuation is similar

GAMLj:Trend analysis

- Simple effects trend analysis: We can now look at the parameters of the simple effects analysis

Trends

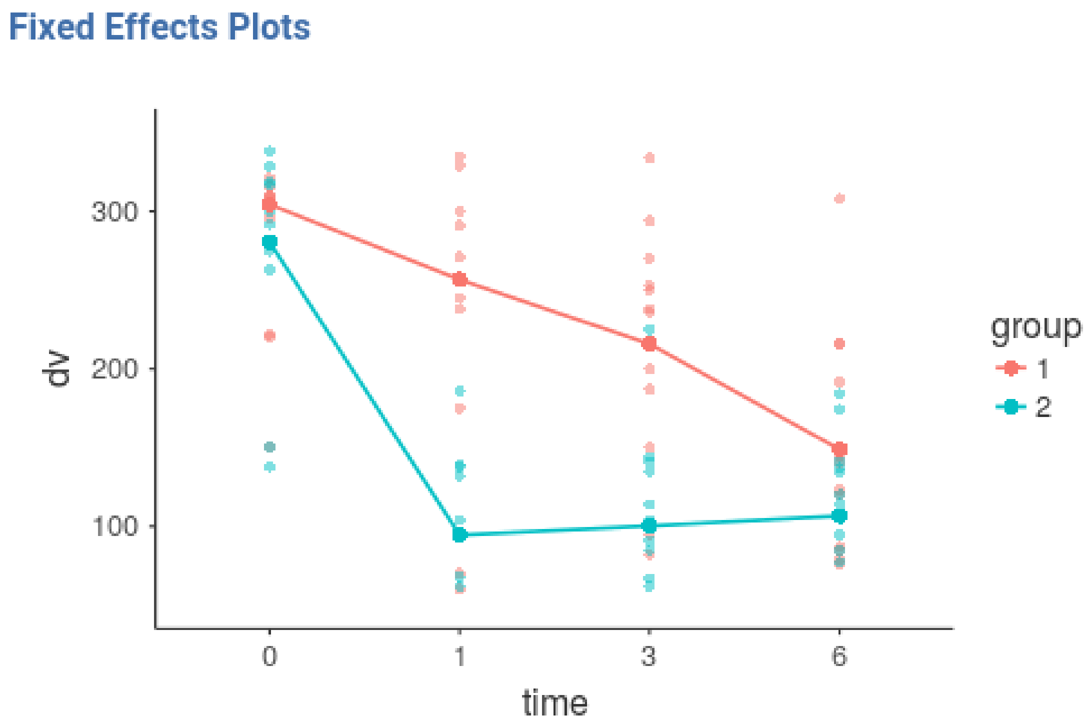
Parameter Estimates for simple effects of time

Moderator				95% Confidence Intervals				
group	Effect	Estimate	SE	Lower	Upper	df	t	p
1	linear	−113.46	15.2	−143.7	−83.2	66.0	−7.481	< .001
	quadratic	−9.62	15.2	−39.9	20.7	66.0	−0.635	0.528
	cubic	−7.32	15.2	−37.6	23.0	66.0	−0.483	0.631
2	linear	−115.25	15.2	−145.5	−85.0	66.0	−7.599	< .001
	quadratic	96.12	15.2	65.8	126.4	66.0	6.338	< .001
	cubic	−42.76	15.2	−73.0	−12.5	66.0	−2.820	0.006

- In group 1 there’s only a linear trend
- In group 2 all three trend are there

GAMLj:Trend analysis

- Simple effects trend analysis: We can now interpret the parameters of the simple effects analysis



- In group 1 there's only a linear trend
- In group 2 all three trends are there

Non-linear effects of time:
The Individual Growth Model

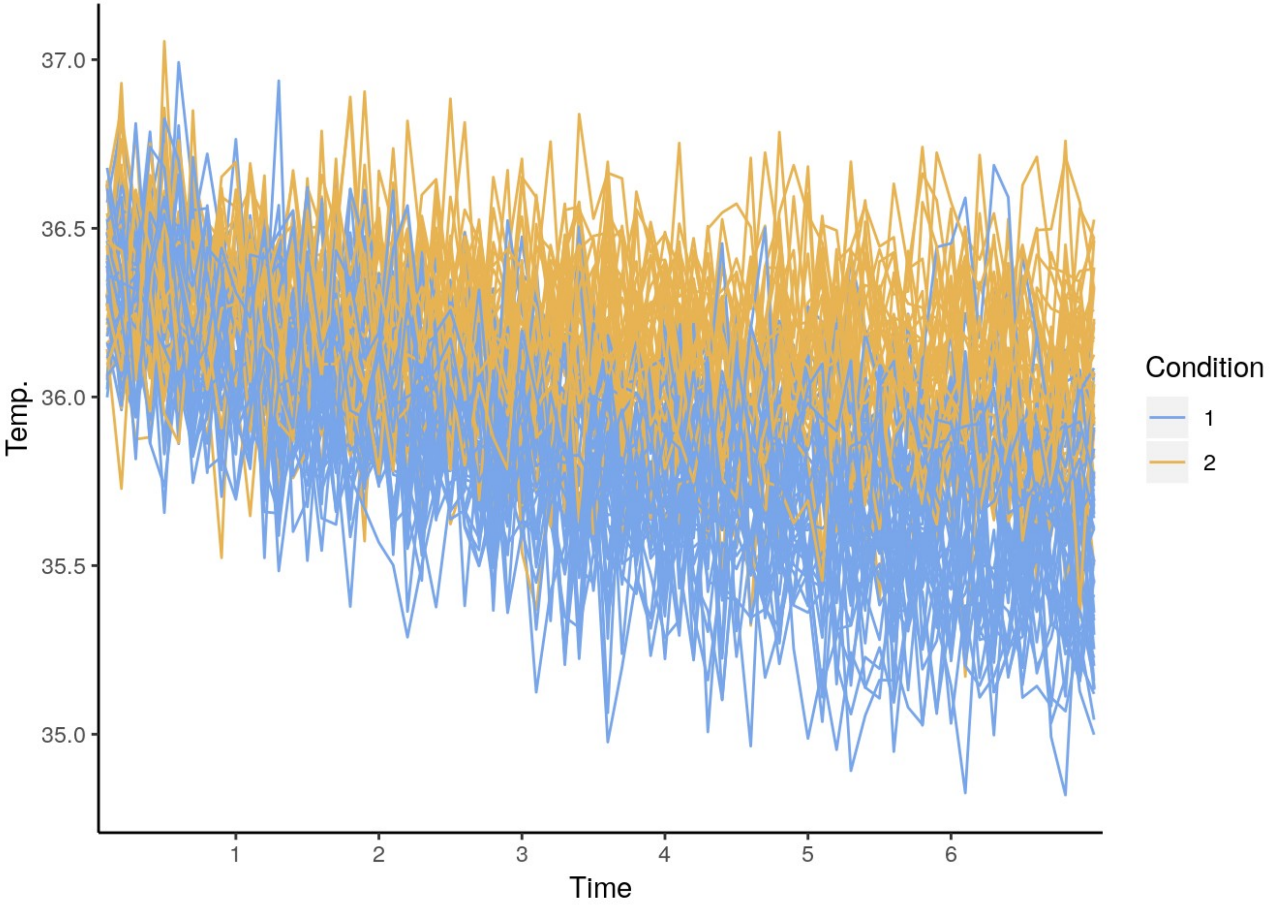
Example

- Imagine the following scenario. Two procedures to decrease the body temperature of the experiment participants are tested. The two procedures are administered between-subjects to two different groups. Let’s identify them as condition 1 and condition 2. Each participant’s body temperature is measured for about 6 minutes every 6 seconds. Time is recorded in tenth of minutes (6 seconds=0.1, 12 seconds=0.2, etc.). We wish to establish whether the temperature decreases over time and if the decrease is different in the two conditions.

Data: temptime.csv

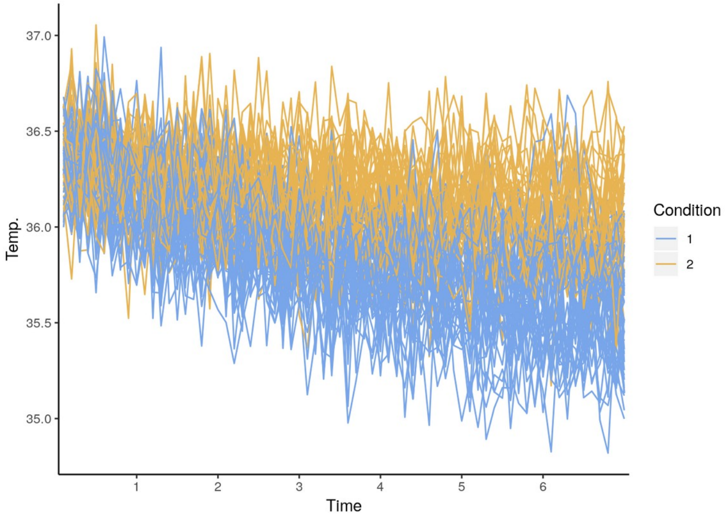
Example

- The data



Charateristics of the design

- It is a repeated-measure design
- The repeated-measure variable is continuous (time)
- The effects cannot be only linear (people will die!!)
- Every participant may show a different pattern of decrease in termp



Charateristics of the design

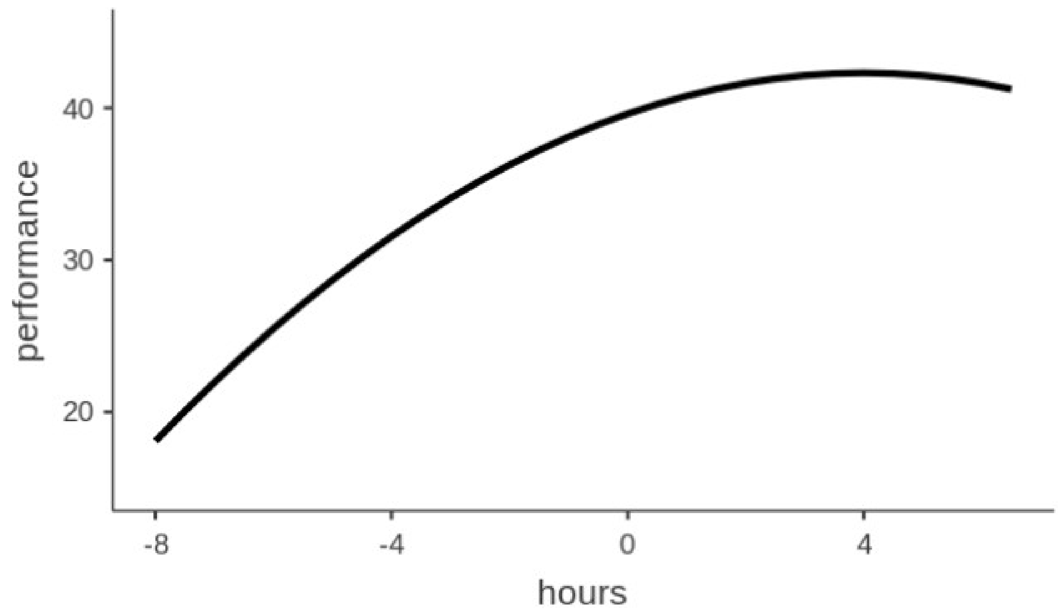
- It is a repeated-measure design: **mixed model**
- The repeated-measure variable is continuous (time): **no problem**
- The effects cannot be only linear: **we need quadratic (or higher) effects**
- Every participant may show a different pattern of decrease in termp: **each effect should be let random across participant**

$$\hat{y}_{it} = a + b_{1i} \cdot x_{it} + b_{2i} \cdot x_{it}^2 + \ldots$$

Curvilinear effects

- In any linear model, trends can be estimated also for continuous independent variables effects

Plots

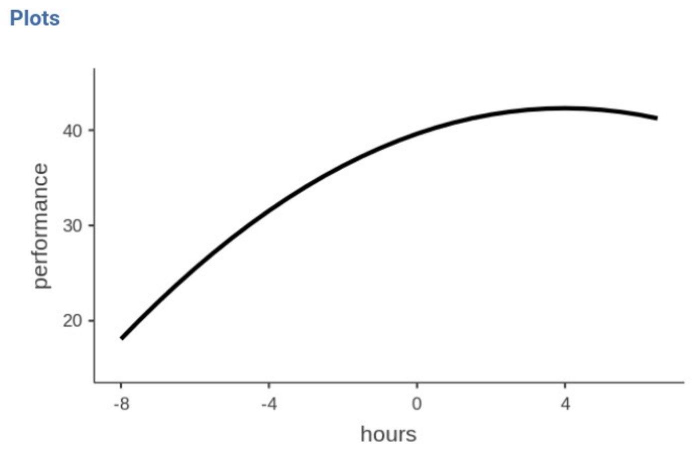


Curvilinear effects

- Trends are added to the model as powers of the independent variable

$$\hat{y} = a + \underbrace{b_1 \cdot x}_{\text{linear}} + \underbrace{b_2 \cdot x^2}_{\text{quadratic}} + \underbrace{b_3 x^3}_{\text{cubic}} \ldots$$

Each power represents a trend as in the polynomial trend analysis

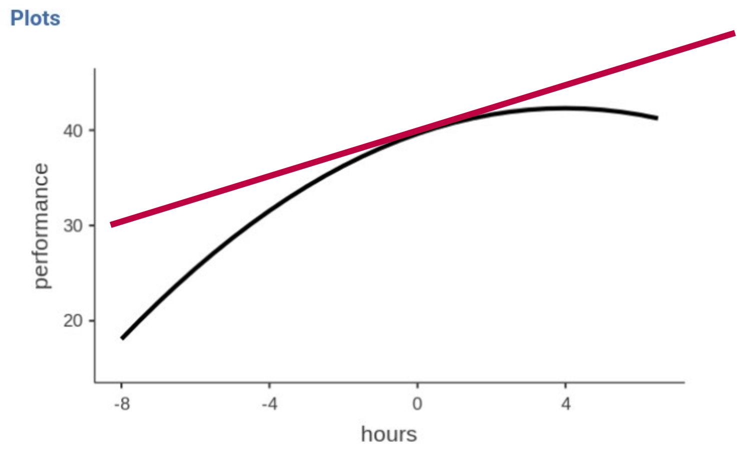


Curvilinear effects

- Trends are added to the model as powers of the independent variable

$$\hat{y} = a + \underbrace{b_1 \cdot x}_{\text{linear}} + \underbrace{b_2 \cdot x^2}_{\text{quadratic}} + \underbrace{b_3 x^3}_{\text{cubic}} \ldots$$

To interpret the results, variables should be centered: remember the interaction!



Curvilinear effects

- We can evaluate the presence of a particular trend by checking the effect size (and significance) of the coefficient associated with it

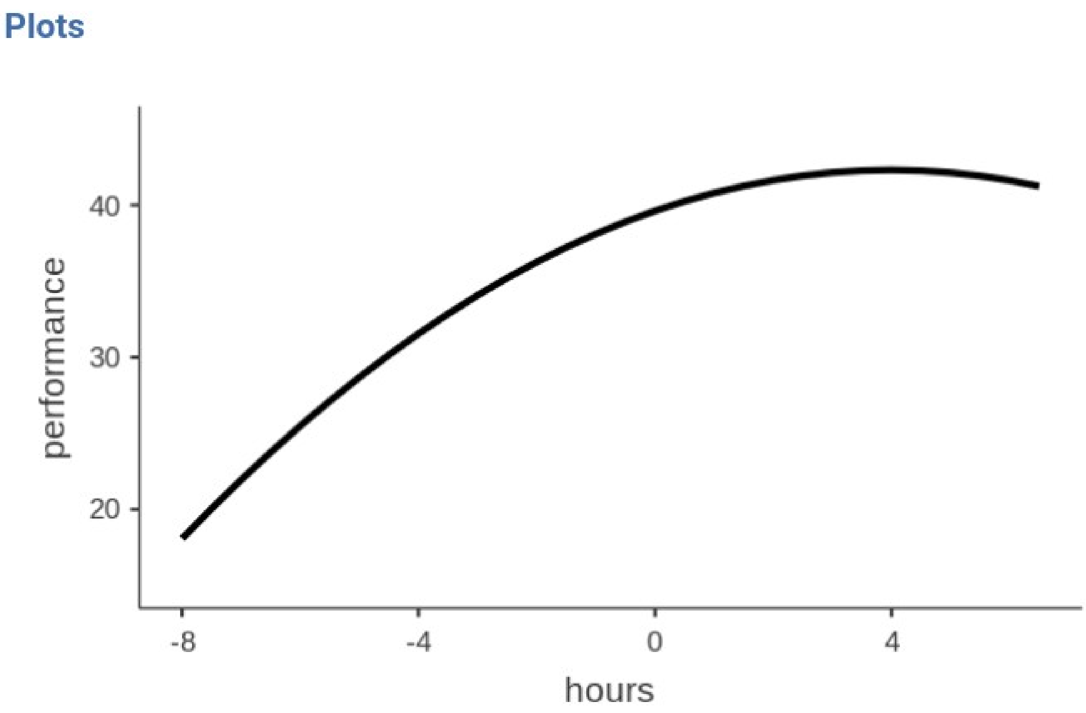
$$\hat{y} = a + b_1 \cdot x + b_2 \cdot x^2 + b_3 x^3 \ldots$$

Fixed Effects Parameter Estimates

			95% Confidence Interval					
Names	Estimate	SE	Lower	Upper	β	df	t	p
(Intercept)	39.62349	0.6639	38.3056	40.9413	0.0000	96	59.683	< .001
hours	1.39841	0.3102	0.7827	2.0141	0.6411	96	4.508	< .001
hours²	−0.16913	0.0409	−0.2502	−0.0880	−0.2504	96	−4.139	< .001
hours³	−0.00212	0.0104	−0.0227	0.0185	−0.0101	96	−0.204	0.839

Curvilinear effects

- This is particular important when we are dealing with effects of time, because the effects of time are seldom linear



Building the model

We translate this in the standard mixed model

$$\hat{y}_{it} = a + b_{1i} \cdot x_{it} + b_{2i} \cdot x_{it}^2 + \bar{b}_{1i} \cdot x_{it} + \bar{b}_{2i} \cdot x_{it}^2$$

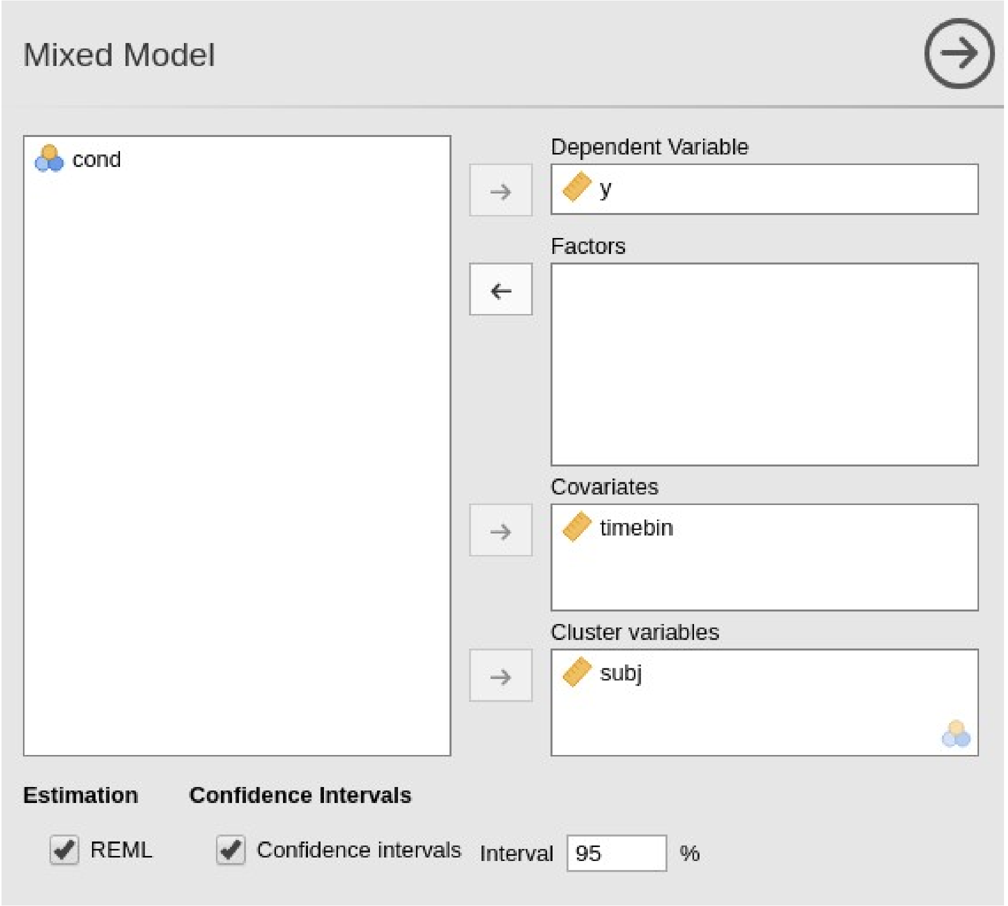
Random coefficients

Fixed coefficients

- Fixed effects? Intercept and linear and the quadratic effects of time
- Random effects? Intercept and linear and the quadratic effects of time
- Clusters? participants

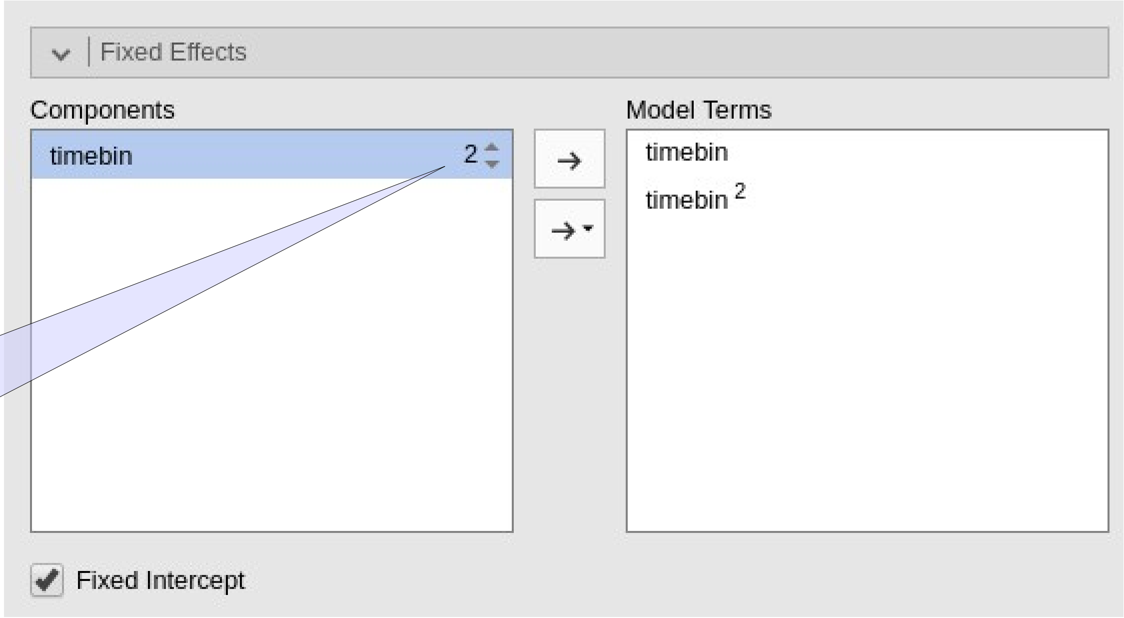
Jamovi GAMLj

- Variables role setup



Jamovi GAMLj

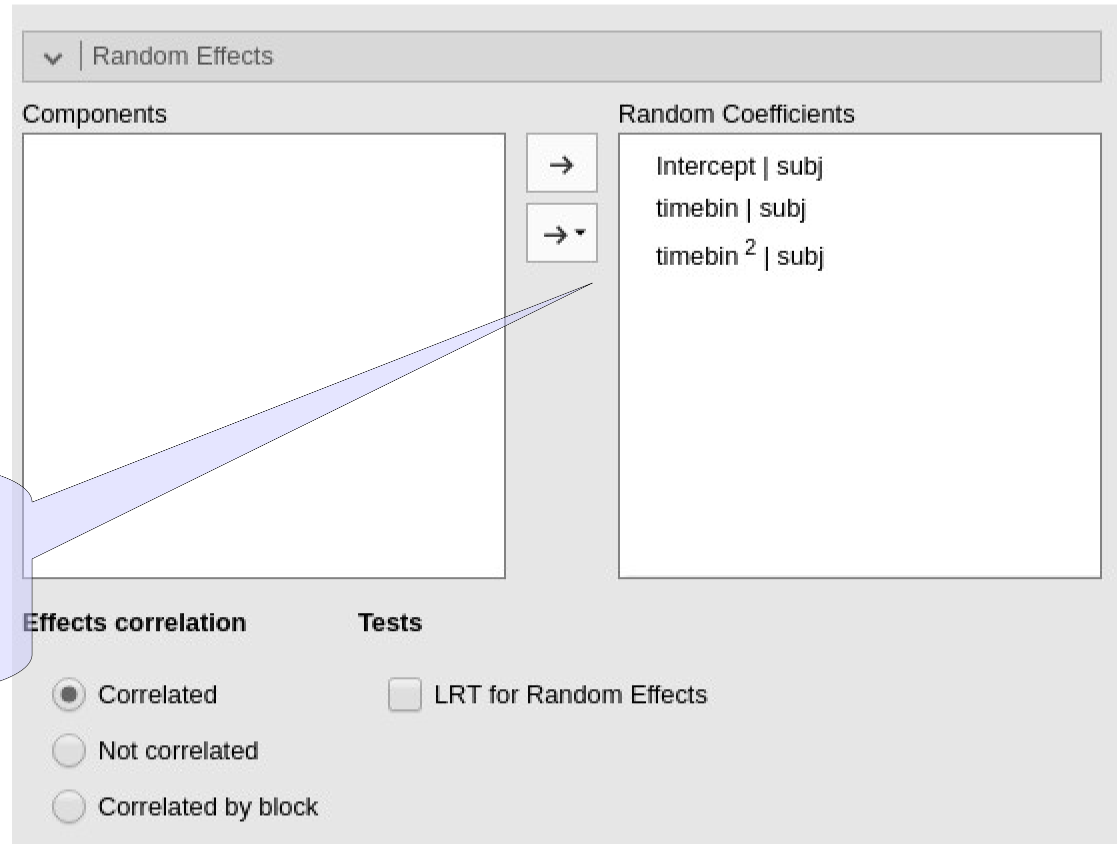
- Fixed effects



We change the order of the effect here

Jamovi GAMLj

- Random effects



Each participant can have his own curve (linear+quadratic)

Jamovi GAMLj

- Random coefficients variances

Random Components

Groups	Name	SD	Variance	ICC
subj	(Intercept)	0.19480	0.03795	0.439
	timebin	0.04104	0.00168	
	timebin²	0.00696	4.84e-5	
Residual		0.22031	0.04854	

Note. Number of Obs: 5600 , groups: subj , 80

Each trend shows some variance, so it was good to let them vary

Jamovi GAMLj

- Fixed coefficients (recall the IVs are centered)

On averge, temp goes down

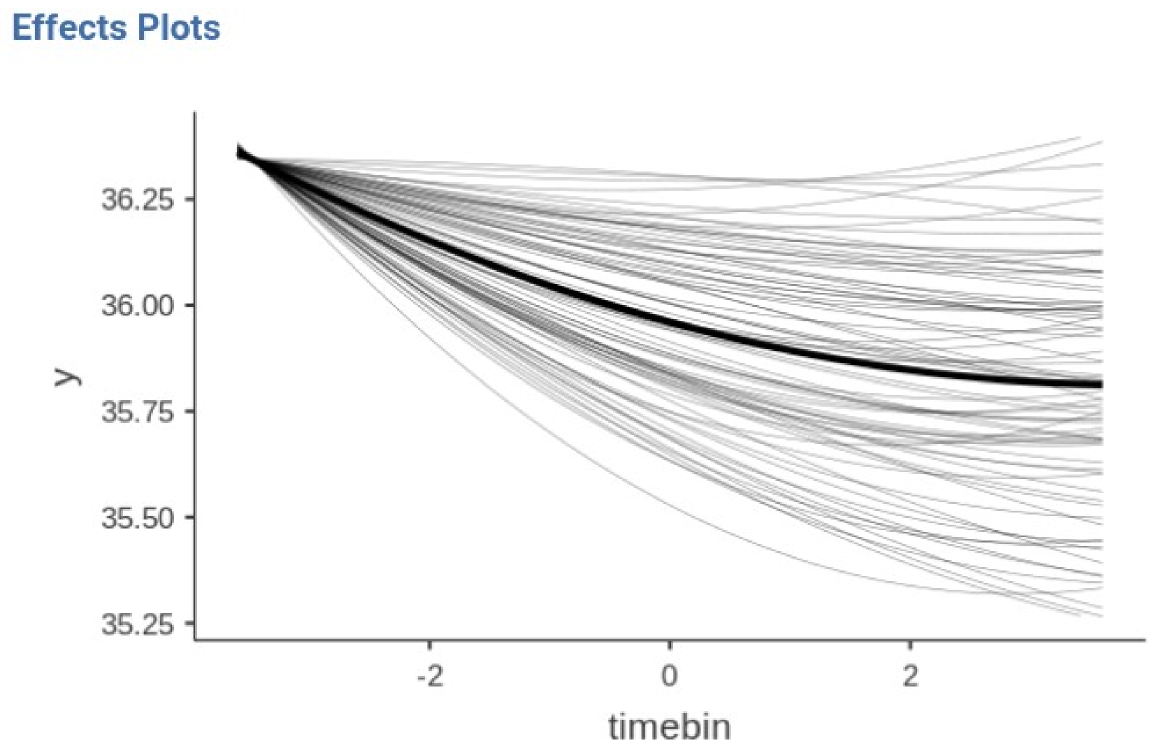
Fixed Effects Parameter Estimates

			95% Confidence Interval				
Names	Estimate	SE	Lower	Upper	df	t	p
(Intercept)	35.95982	0.02222	35.91627	36.0034	79.0	1618.15	< .001
timebin	−0.07630	0.00481	−0.08573	−0.0669	79.1	−15.85	< .001
$timebin^2$	0.00984	0.00112	0.00764	0.0120	88.1	8.78	< .001

On averge, temp decline smooth out: becomes flatter

Jamovi GAMLj

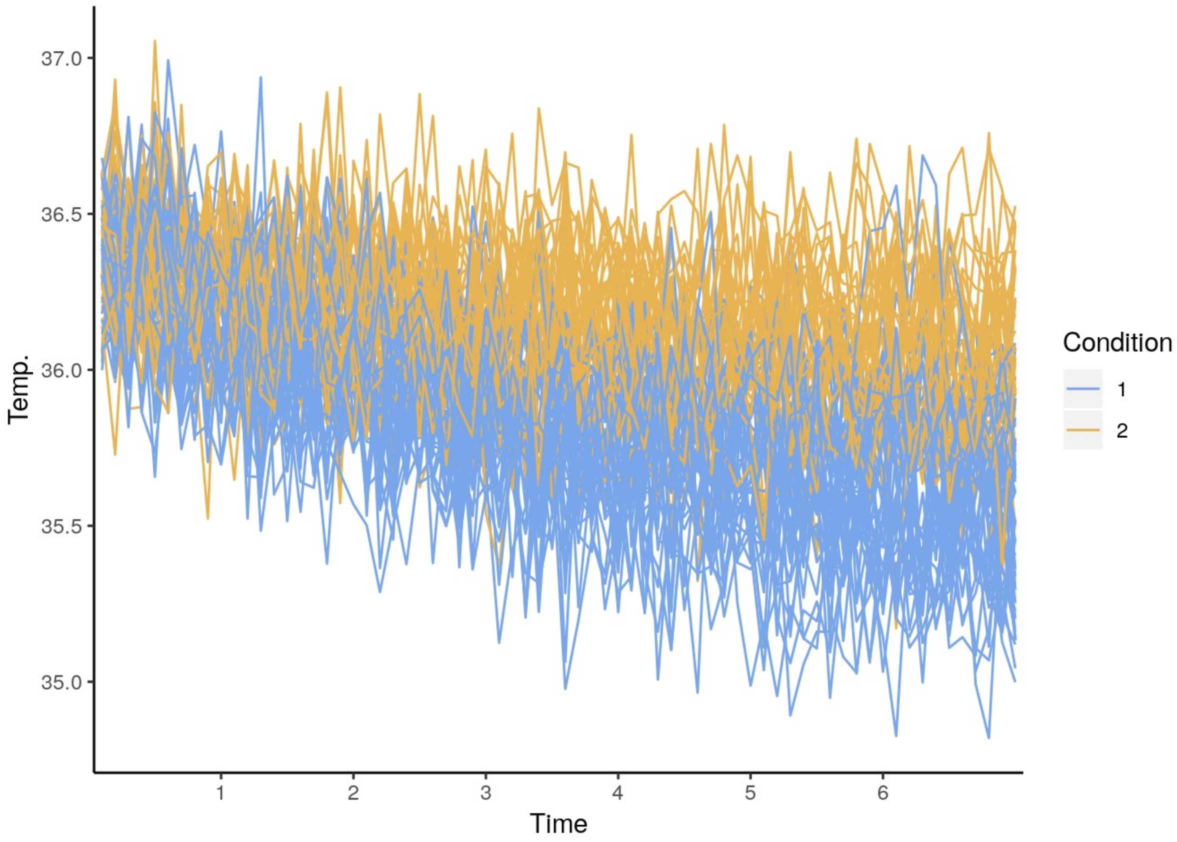
- Plot (fixed and random effects)



The Individual Growth Model

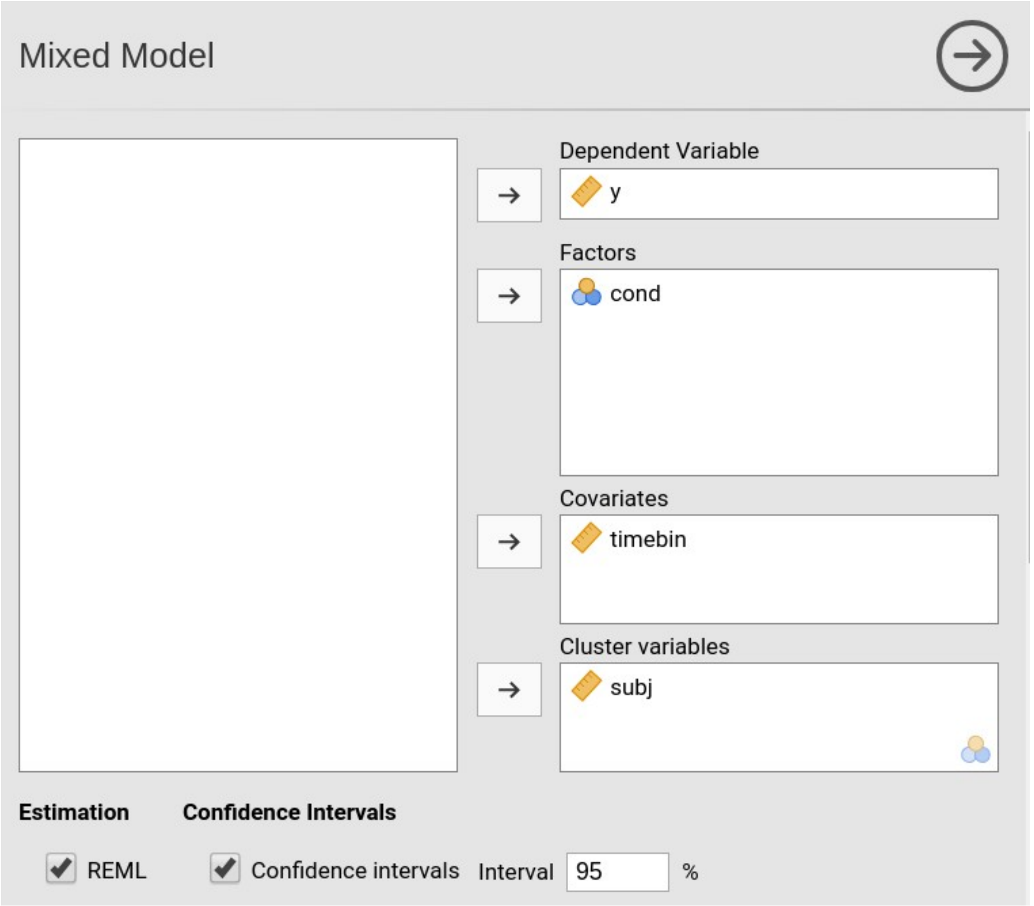
Establish the effect of condition

- Can we say that the two groups have different trends



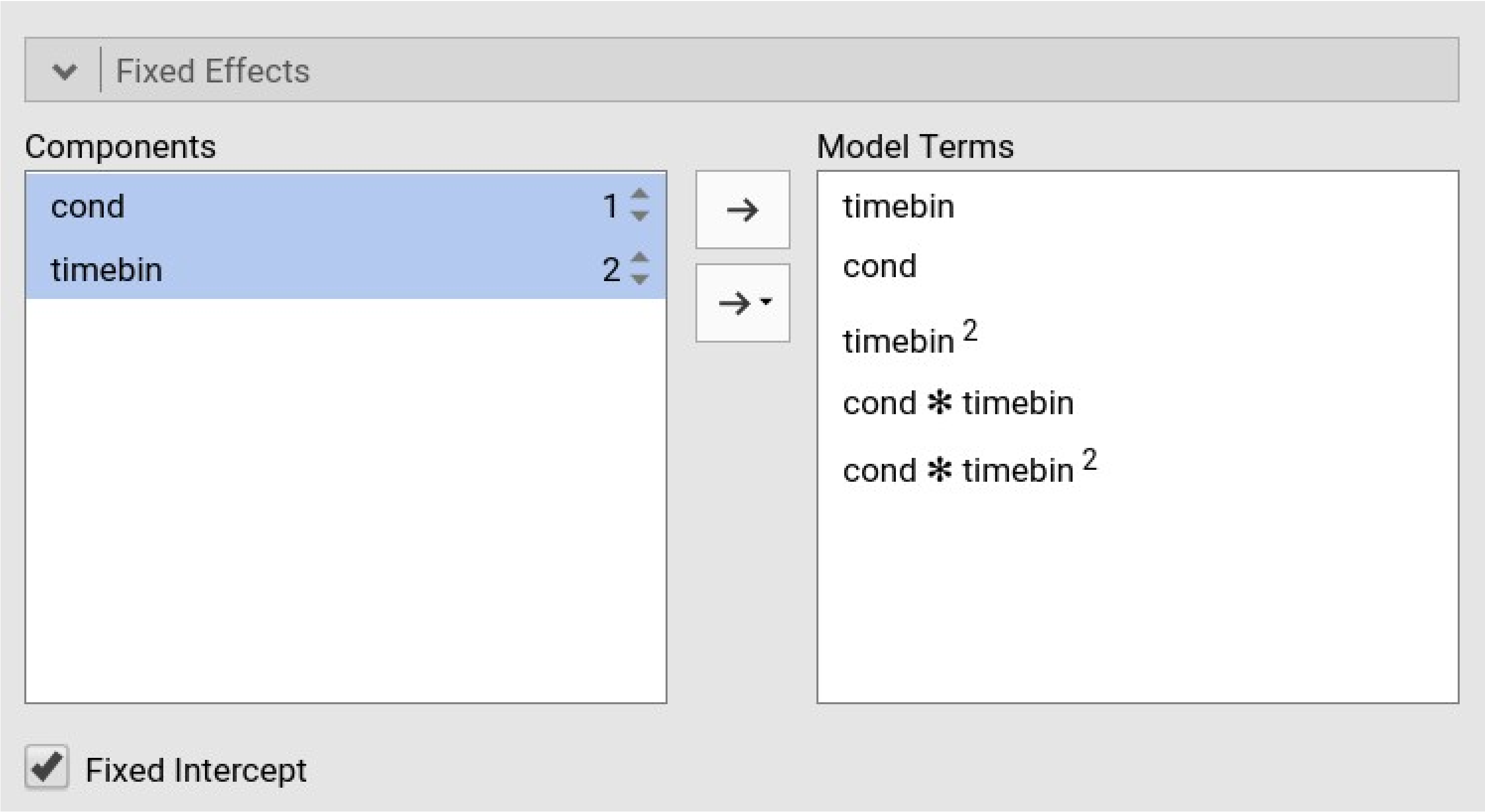
Establish the effect of condition

- Add the condition as a factor



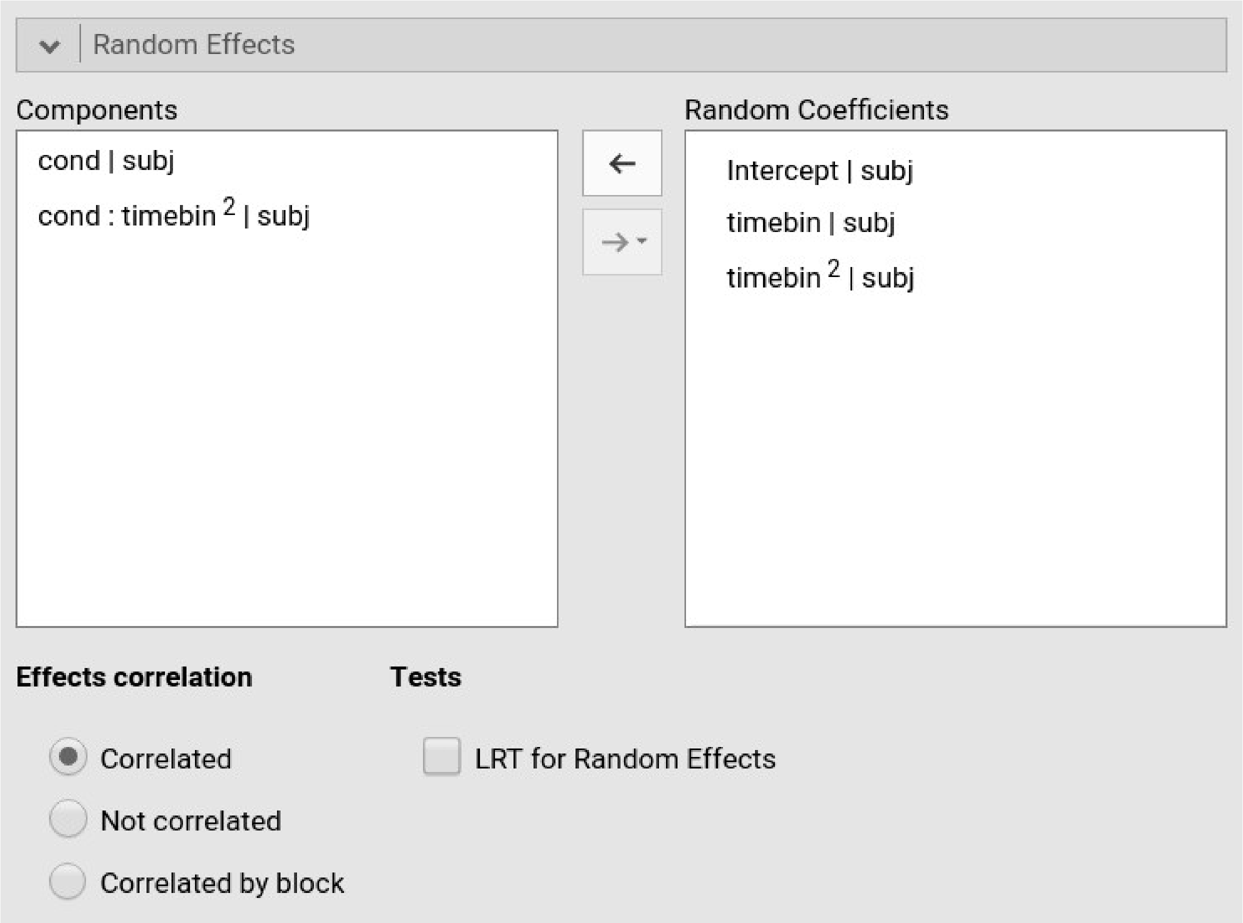
Establish the effect of condition

- Put in the model its main effects and interactions with the time effects



Establish the effect of condition

- Do not change the random effects: condition is not repeated measure factor, so its effects do not vary from subject to subject



Establish the effect of condition

- Results

Main effect of "cond": the average temp is different in the two groups

Fixed Effects Parameter Estimates

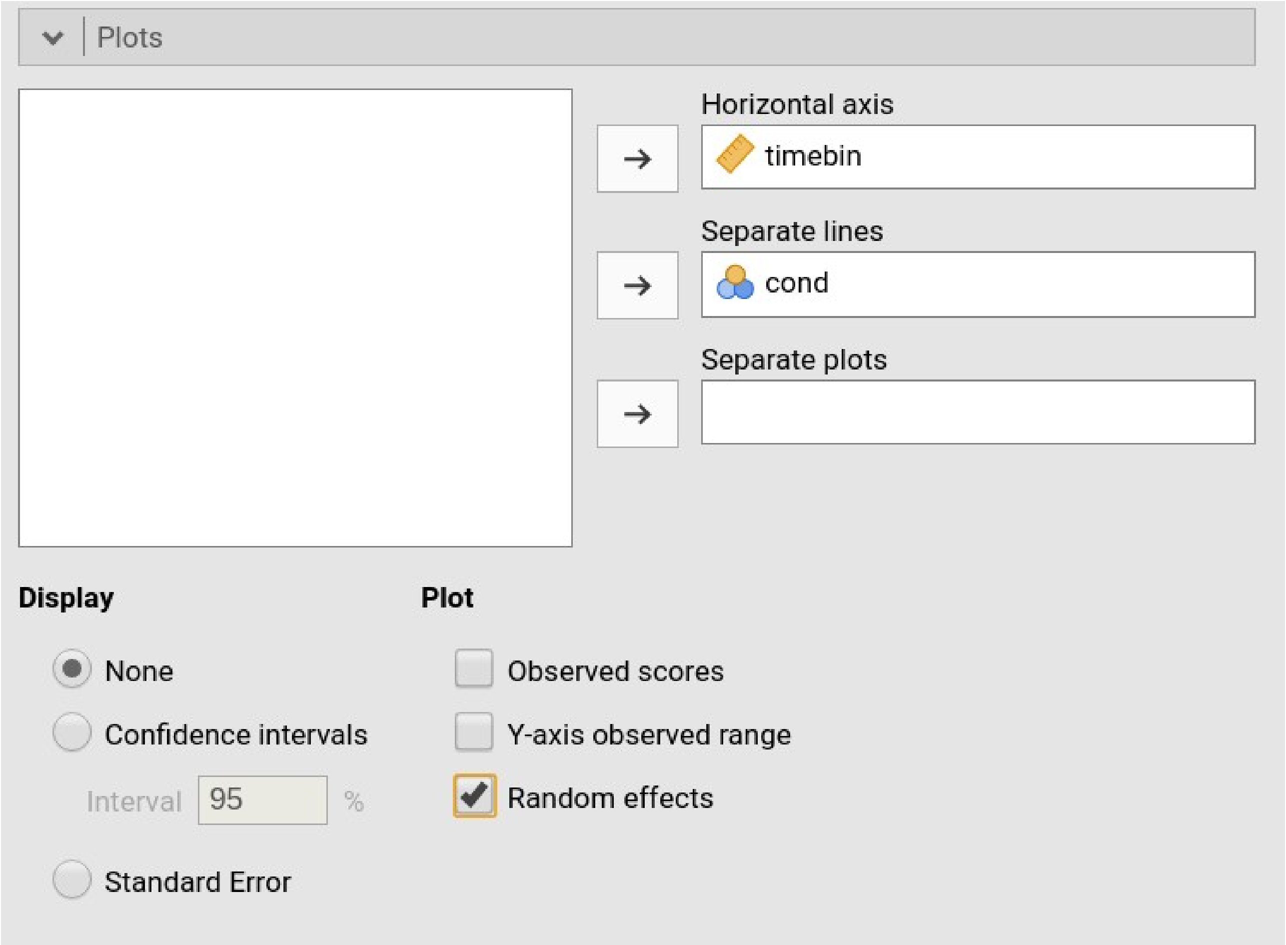
Names	Effect	Estimate	SE	95% Confidence Interval Lower	95% Confidence Interval Upper	df	t	p
(Intercept)	(Intercept)	35.97249	0.01171	35.94954	35.99545	78.1	3071.32	< .001
timebin	timebin	−0.07404	0.00346	−0.08083	−0.06725	78.5	−21.37	< .001
cond1	2 - 1	0.33786	0.02342	0.29195	0.38378	78.1	14.42	< .001
timebin²	timebin²	0.00943	9.42e−4	0.00758	0.01127	102.3	10.01	< .001
timebin ✻ cond1	timebin ✻ 2 - 1	0.06014	0.00693	0.04656	0.07372	78.5	8.68	< .001
cond1 ✻ timebin²	2 - 1 ✻ timebin²	−0.01105	0.00188	−0.01474	−0.00736	102.3	−5.87	< .001

Interaction cond*linear: the two groups show different linear trends

Interaction cond*quadratic: the two groups show different curvatures

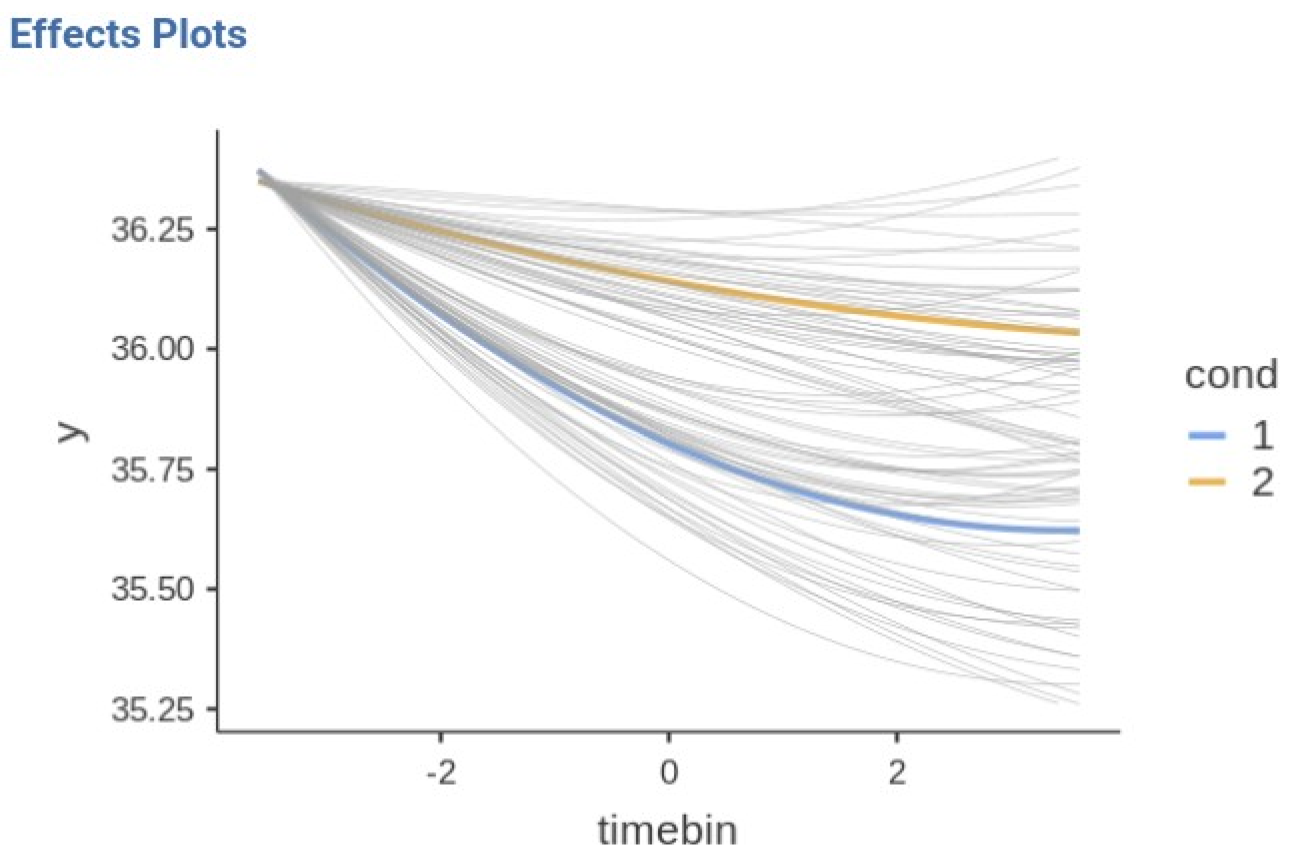
Establish the effect of condition

- plot



Establish the effect of condition

- plot



Generalized Mixed Models

Generalized Linear Models

- There are many situations where the dependent variable is not normally distributed:

  - Predicting groups

  - Predicting choices (yes/no, left/right, etc.)

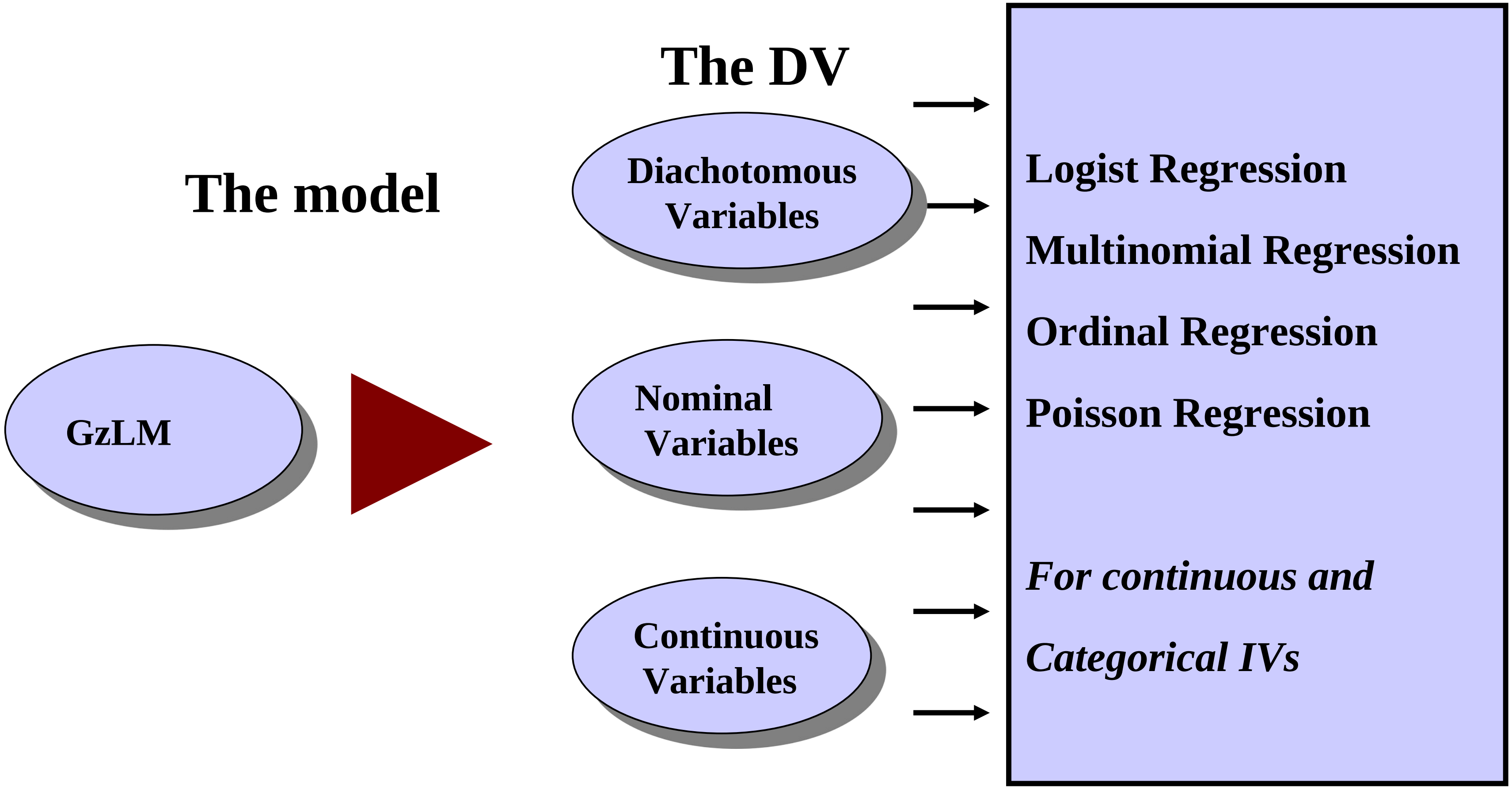
  - Predicting frequencies of behavior

GLM

When the assumptions are NOT met because the dependent variable is not normally distributed (dichotomous, frequencies, categorical etc), we generalize the GLM to the

Generalized Linear Model (GzLM)

Generalized Linear Model



GGLM

- The generalized linear model is a linear model with the dependent variable modelled with a specific function (link function) and with specific error distribution

Generalized Linear Model

$$f(y_i)=a+b_1\cdot x_{1i}+b_2\cdot x_{2i}+..b_k\cdot x_{ki}+e_i$$

Dependent variable

Specify a distributio n shape

Generalized mixed model

- Applying this logic we obtain a large set of possible statistical techniques

$$f(y_i) = a + b_1 \cdot x_{1i} + b_2 \cdot x_{2i} + \ldots b_k \cdot x_{ki} + e_i$$

Dependent Variable	**Model**
Continuous	**Linear**
Dichotomous	**Logistic**
Categorical	**Multinomial**
Ordinal	**Ordinal**
Frequencies	**Poisson**

Logic

- When the DV is categorical, we can use the **Generalized Linear model** which allows to apply regression/ANOVA tecniques to categorical dependent variables

- For repeated measures (or in general dependent data), we use the **Generalized Linear Mixed model** to allow coefficients to vary randomly across clusters, thus taking dependency into the account

jamovi GAMLj

- Imagine a study conducted in 70 schools. In each school the same exam is taken by students of equivalent age and grade. For each student, we recorded whether the student passed the exam, pass, the student's score in math test, math, and the number of extracurricular activities the student undertook during the semester.

- The researcher wants to estimate the effect of the math test on the probability of passing the exam, together with the amount of extracurricular activities may moderate the math effect.

- Each school has a different number of students, ranging from 51 to 100. Each student presents three values: the score in the math test, the number of activity undertaken and whether the exam was passed pass=1 or not, pass=0.

Design

- Schools are the clusters

Because we have a dichotomous dependent variable, we need a logistic regression

Because we have clustered data, we need a logistic mixed regression

Frequencies

Frequencies of pass

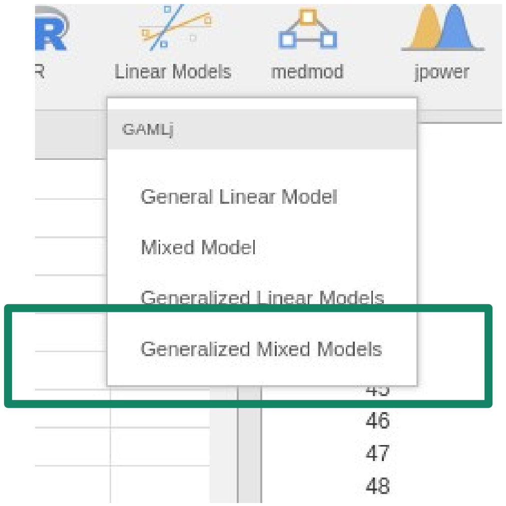
Levels	Counts	% of Total	Cumulative %
0	2479	49.2 %	49.2 %
1	2562	50.8 %	100.0 %

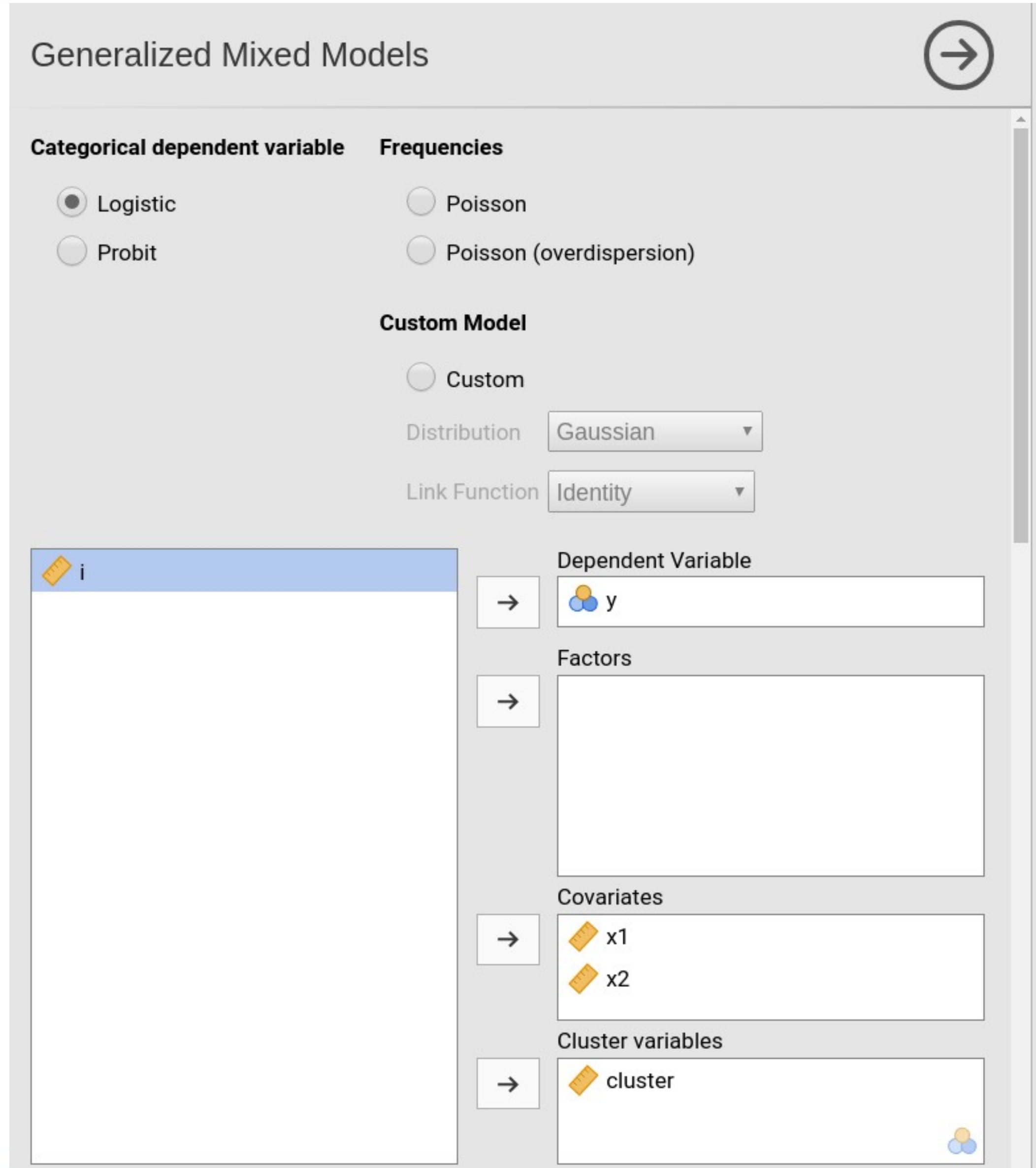
Frequencies of school

Levels	Counts	% of Total	Cumulative %
1	95	1.9 %	1.9 %
2	62	1.2 %	3.1 %
3	60	1.2 %	4.3 %
4	56	1.1 %	5.4 %
5	90	1.8 %	7.2 %
6	72	1.4 %	8.6 %
7	82	1.6 %	10.3 %
8	89	1.8 %	12.0 %
9	100	2.0 %	14.0 %
10	59	1.2 %	15.2 %

GzLMM

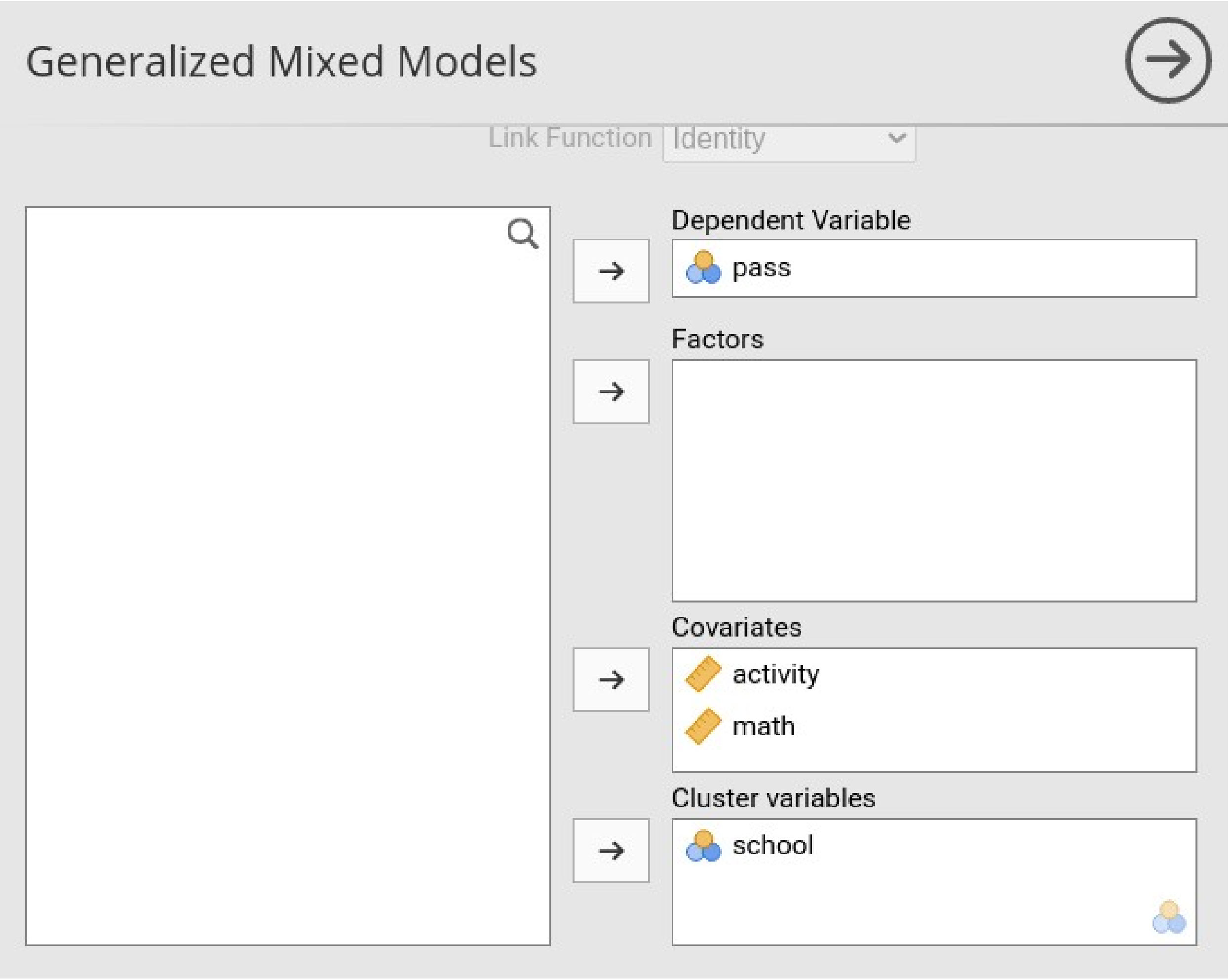
- We launch the module





GzLMM

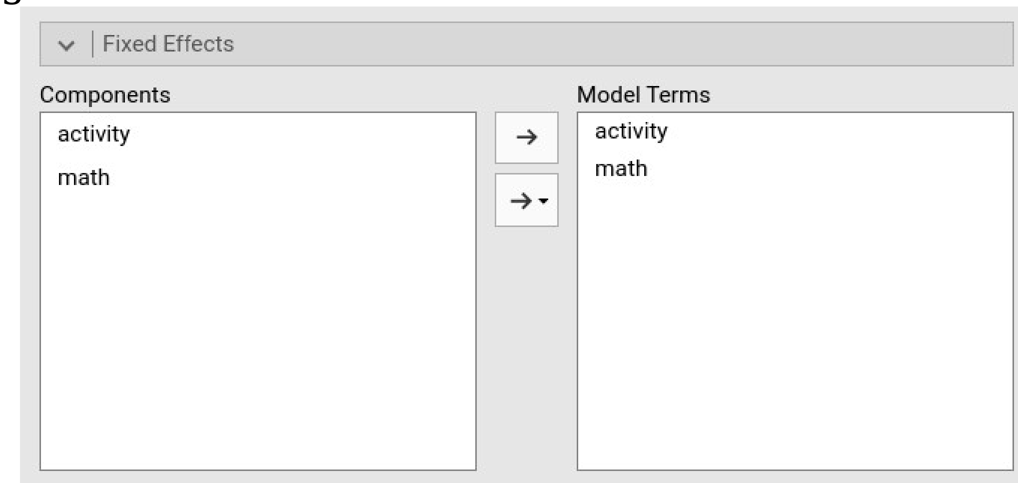
- We select the variables role



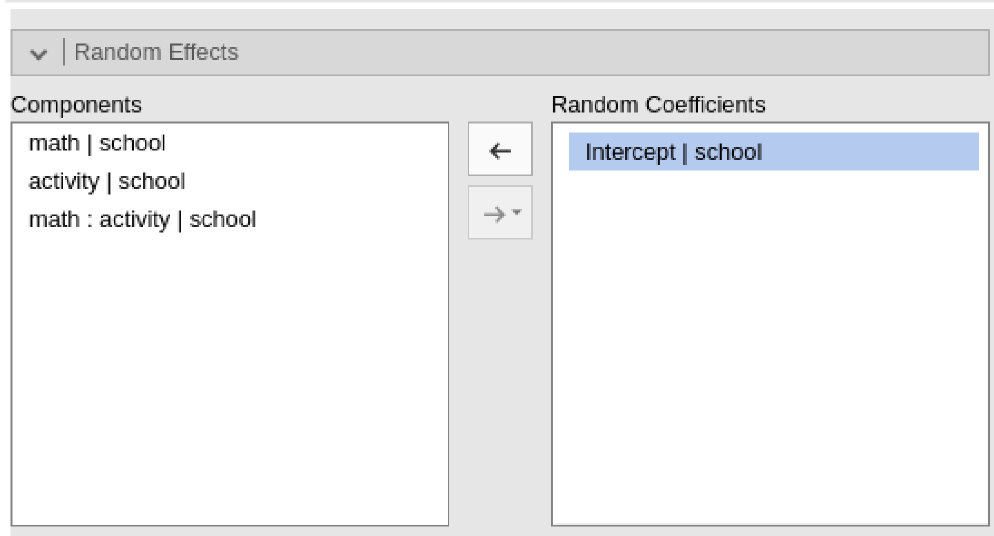
GzLMM

- Define the model parameters

Fixed Effects



Random Effects



Results

- Info table

Direction of the model: What are we predicting?

R-squared for the whole model and for the fixed effects

Model Info

Info	Value	Comment
Model Type	Logistic	Model for binary y
Call	glm	pass ~ 1 + math + activity + math:activity + (1 | school)
Link function	Logit	Log of the odd of y=1 over y=0
Direction	P(y=1)/P(y=0)	P(pass = 1) / P(pass = 0)
Distribution	Binomial	Dichotomous event distribution of y
LogLikel.	−2785.0640	Less is better
R-squared	0.0395	Marginal
R-squared	0.3787	Conditional
AIC	5580.1300	Less is better
BIC	5612.7547	Less is better
Deviance	5287.0900	Conditional
Residual DF	5036.0000	

[3]

Results

- Random component

Random Components

Groups	Name	SD	Variance	ICC
school	(Intercept)	1.32	1.75	0.347
Residuals		1.00	1.00	.

Note. Number of Obs: 5041 , groups: school 70

Proportion of variance accounted for by the intercepts

Results

- Fixed effects: **Omnibus Tests**

Fixed Effect Omnibus tests

	X^2	df	p
activity	29.4	1.00	< .001
math	91.8	1.00	< .001

GAMLj uses the Chi-Squared

Results

- Fixed effects: **coefficients**

Here we found the exp(B)

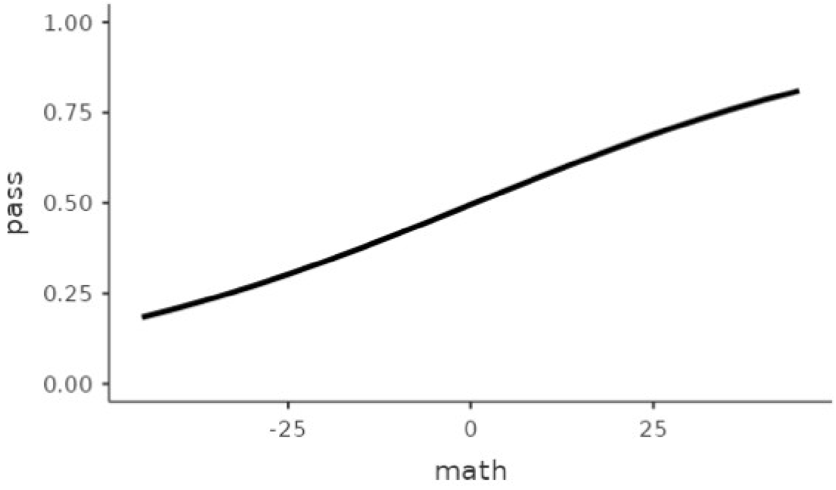
Fixed Effects Parameter Estimates

Names	Estimate	SE	exp(B)	95% Exp(B) Confidence Interval Lower	95% Exp(B) Confidence Interval Upper	z	p
(Intercept)	−0.0188	0.16207	0.981	0.714	1.348	−0.116	0.908
activity	−0.1764	0.03255	0.838	0.786	0.894	−5.419	< .001
math	0.0327	0.00341	1.033	1.026	1.040	9.579	< .001

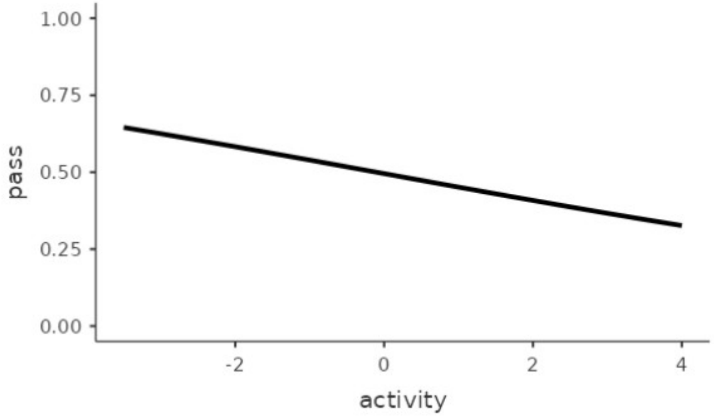
Results

- Plot: The probability to pass the exam as a function of the independent variables

Effects Plots



Effects Plots



Logic

- **General linear model** allows for analyzing a variety of design with normally distributed DV by apply regression/ANOVA tecniques

- For repeated measures (or in general dependent data), we use the **Linear Mixed model** to allow coefficients to vary randomly across clusters, thus taking dependency into the account

- When the DV is categorical, we can use the **Generalized Linear model** which allows to apply regression/ANOVA tecniques to categorical dependent variables

- For repeated measures (or in general dependent data), we use the **Generalized Linear Mixed model** to allow coefficients to vary randomly across clusters, thus taking dependency into the account

The end